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Multi-market optimization of flexible hydropower

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Preface

This project report concludes the work conducted throughout the course *TIØ4500 - Managerial Economics and Operations Research, Specialization Project* at the Norwegian University of Science and Technology, Department of Industrial Economics and Technology Management.

We extend our sincere gratitude to Stian Backe, whose guidance and expertise have been invaluable throughout this project. His insight and experience within energy and power market modelling have greatly supported us in navigating complex challenges. We would also like to express our appreciation to Kasper Thorvaldsen for his valuable discussions, constructive feedback, and helpful insights. Finally, we would like to thank Anders Strand Ryssdal and Ivar Refsdal for their consistent contributions and valuable input to the work presented in this specialization project.

Trondheim, December 2025

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Abstract

In an evolving power system, balancing services are expected to become an increasingly important consideration. A growing share of intermittent generation technologies, increased interconnection capacities, and rising electricity demand are among the key drivers behind this development. Furthermore, due to the inherent structure of power markets, flexible resources such as hydropower are well-positioned to participate across multiple markets. These aspects provide a strong motivation for investigating how flexible hydropower can be operated more efficiently across multiple markets, not only to increase profitability but also to ensure system stability. As many hydropower plants are ageing and are currently being considered for upgrades, the question of whether additional investments in capacity upgrades of hydropower plants can help address the challenges mentioned above.

The project analyses the various electricity and capacity markets with respect to both their technical characteristics and economic incentives. This is followed by a literature review revealing a research gap on flexible hydropower plants participating in a detailed multi-market framework. Thus, the following research problem has been identified:

Optimization of flexible hydropower operation by incorporating multi-market characteristics and relevant modelling considerations to quantify operational performance and revenues.

This research is addressed through a deterministic optimisation model with the objective of maximising operational revenues across energy and several capacity markets. A case study of a hydropower producer located in the NO2 price area is conducted to assess whether multi-market participation can contribute to justifying a potential capacity upgrade of the power plant.

From an operational perspective, the results indicate that participation across both energy and reserve markets, without the consideration of activation, leads to less aggressive plant operation. From an economic perspective, participation in capacity markets results in increased operational revenues, as the water resource is utilized more effectively. The results further indicate, with the underlying assumptions and input data employed, that a capacity upgrade could be economically justified. With further development, the findings also suggest that the proposed model may evolve into a more robust decision-support tool for evaluating future investments.

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1 Introduction

Despite sufficient reservoir levels in Norwegian hydropower reservoirs and relatively moderate outdoor temperatures on the 12th of December 2024, power prices surged to 13.37 $\frac{NOK}{kWh}$ (Andersen, 2024; NVE, 2025a). This price level is almost twenty times higher than the annual mean in 2024, illustrating the significant price fluctuations currently occurring in the market (SSB, 2025a).

In the power system, electricity must be produced and consumed simultaneously to maintain the system frequency. A range of electricity markets has therefore been developed to ensure operational feasibility. While long-term financial instruments aim to provide predictability, the day-ahead (DA) market provides operational guidance one day before physical delivery. The market-clearing price (MCP) is determined by the intersection of the supply and demand curves, established through bids placed according to the merit-order principle. Furthermore, the intraday (ID) market functions as a continuous bilateral trading arena, allowing participants to adjust their scheduled positions. Closer to real time, several reserve markets are activated to counteract imbalances and maintain a stable frequency.

Theoretically, the MCP equals the bid of the marginal unit in the merit order required to satisfy demand. In the European market, this marginal unit is usually a gas-fired power plant (Blume-Werry et al., 2021). Although hydropower typically has very low marginal production costs, this is not reflected in its supply bids. On the one hand, reservoir-based hydropower can defer production to future periods when electricity is scarcer, creating an opportunity cost associated with alternative uses of stored water. Furthermore, water in the reservoir functions as a societal safeguard against costly blackouts, an aspect that must be incorporated into bidding strategies. This opportunity cost is commonly referred to as the water value, which forms the basis for the hydropower marginal cost bidding. Moreover, one unit of hydropower generation can effectively displace the marginal gas-fired unit that would otherwise set the price in the continent, given available transfer capacity. Consequently, the severe constraints on European gas supply following Russia's invasion of Ukraine triggered an energy crisis, ultimately driving power prices to notably high levels.

Achieving the target of net-zero greenhouse gas emissions by 2050 requires substantial innovation and regulatory measures within the energy sector. Instruments such as the EU Emissions Trading System (ETS) and rapid cost reductions in wind and solar technologies have made variable renewable energy sources (VRES) an increasingly important part of the energy mix, as illustrated in Figure 1 and Figure 2a. At

the same time, consumers have altered their consumption patterns through the widespread adoption of electric vehicles, as shown in Figure 2b. Together, these developments increase the risk of volatile production and consumption patterns, thereby increasing the risk of imbalances in the power system (Belsnes et al., 2016).

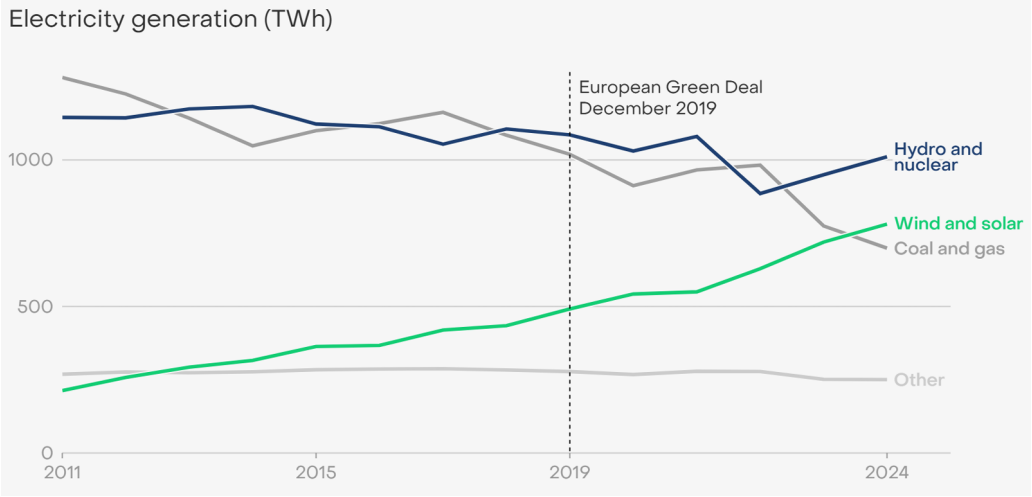
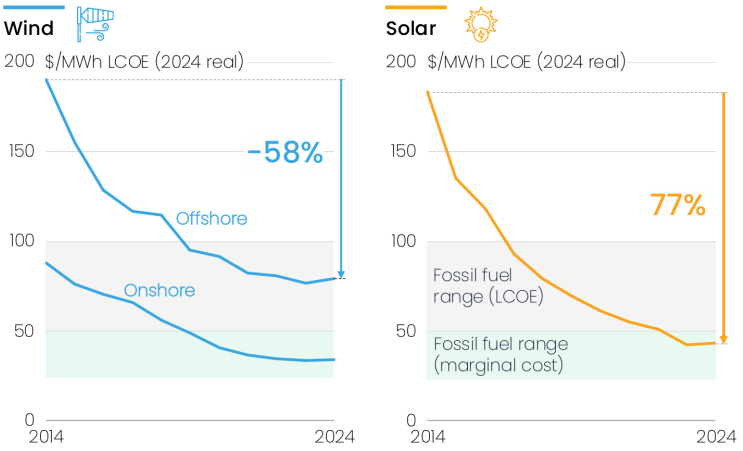
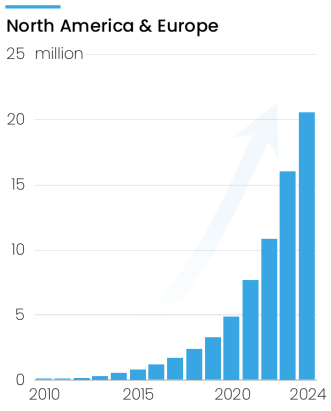


Figure 1: Development of solar and wind power generation compared to other resources in Europe (Ember, 2025b).



(a) Cost improving VRES



(b) Increased demand for EVs

Figure 2: Cost developments in VRES and increased EV demand (Ember, 2025a).

Norway's power system relies heavily on hydropower, which accounts for approximately 90% of total electricity production. As a weather-dependent resource, hydropower is vulnerable to dry periods that can lead to low reservoir levels, meaning the country cannot always be fully self-sufficient in electricity supply. To mitigate this risk, Norway has over time invested in several interconnectors to neighbouring European countries, enabling more efficient utilisation of electricity across regions (RME, 2025). Increased integration with the European power market, however, also exposes Norway to greater uncertainty in continental power prices.

Interconnecting the European grid reflects broader political and technological changes in the European energy market, necessitating increased flexibility. Mentens et. al. (2025) emphasize that such flexibility is essential for effective grid management and a reliable power supply. They distinguish three overarching mechanisms: i) spatial adjustment of demand and supply through interconnectors, ii) demand-response systems facilitated by price signals, and iii) temporal adjustment of supply and demand through storage solutions. Temporal flexibility may be provided by batteries, prosumers, or upgrades that enhance the flexibility of hydropower plants. However, uncertainty in battery deployment and future load development favours reservoir-based hydropower as a stabilizing force in the market (Statnett, 2025b). Warland and Mo (2016) argue that flexible, fast-responding hydropower plants capable of producing during peak-price periods will benefit most in an increasingly uncertain system. This increases the likelihood of operation at capacity bounds. Consequently, physically accurate modelling of hydropower capacity becomes essential for delivering the flexibility increasingly required by the evolving energy system.

The growth of intermittent generation, increased interconnection, gas shortages, and new policy measures all underline the need to study non-conventional hydropower planning. Belsnes (2016) suggests that hydropower producers may need to allocate a larger share of their capacity toward the intraday and reserve markets. To optimise future production, hydropower producers must therefore schedule watercourses in ways that enable participation in multiple markets simultaneously.

The profitability of providing flexibility in a multi-market setting has already triggered numerous investment plans among Norwegian hydropower producers. Concurrently, simple analyses show that increased hydropower flexibility can significantly reduce power price variability (Messel Larsen, 2024). A clear example is Eviny, a major Norwegian power producer, which plans to double the installed capacity of three hydropower plants to capture additional market value. Although profit is the primary motive, such investments are also expected to alleviate price pressure (Messel Larsen,

2025). Investigating how flexible hydropower, capable of contributing to both price and frequency stability, can be economically optimized is therefore a highly relevant operations research topic for this specialization project.

1.1 Purpose of the project

This project seeks to establish an understanding of the electricity markets available to Norwegian power producers, develop a fundamental comprehension of how hydropower scheduling is modelled within an optimization framework subject to physical constraints, and identify research gaps in the academic literature on hydropower planning. The findings from the literature review indicate that multi-market optimisation models have received limited attention, which has led to the following research questions:

- What are the operational impacts of multi-market participation on the hydro power plant's cost-optimal capacity utilization and ramping hours?
- What is the added value of including several capacity markets in a multi-market optimisation model?
- How costly can capacity upgrades be while remaining economically viable, given the increased earning potential?

The main purpose of this project is therefore to investigate how a deterministic multi-market model allocates the capacity of a single hydropower plant across the available markets, and to assess the added value this may provide to the producer. The results represent an optimistic estimate by construction, but demonstrate that the increased revenue potential may justify the investment costs reported in industry benchmarks. Participation in capacity markets causes the production level to fluctuate more frequently, but typically with smaller amplitudes, and the reservoir is emptied less often due to an overall lower production level. Factors such as the available reserve capacity and the relative price interaction between individual reserve markets and the energy markets are decisive for determining which reserve products contribute most to increasing revenues in an optimization setting. The work presented in this project can serve as a base for future research on more realistic modelling of multi-market participation.

1.2 Outline of the project report

The remainder of the report is organized as follows. Section 2 presents a background on the various power markets with a subsequent assessment of the need for balancing services and the price dynamics across these markets. Section 3 introduces key aspects of hydropower production in Norway and outlines the considerations relevant when modeling a hydropower plant within optimization frameworks. Section 4 provides a qualitative review of the relevant literature and modeling approaches, followed by a problem description in Section 5.

Further, Section 6 presents the mathematical model together with the assumptions and simplifications applied during implementation. This is followed by a short-term model validation in Section 8, aimed at assessing the model's behaviour and decisions at fine temporal resolution. Section 9 presents a computational study evaluating both operational and economic performance, which is subsequently linked to an investment analysis of various capacity-upgrade scenarios. Concluding remarks are provided in Section 10 and future research directions are outlined in Section 11.

2 Power markets

This section presents a brief overview of the Norwegian power system, followed by an introduction to the different segments of the electricity market and a simple analysis of market prices. The markets considered include the spot market, commonly referred to as the day-ahead market, the intraday market, and the various reserve markets. An overview of the main markets discussed in this study is provided in Figure 3. Long-term contracts and financial markets are excluded, as they fall outside the scope of this project.

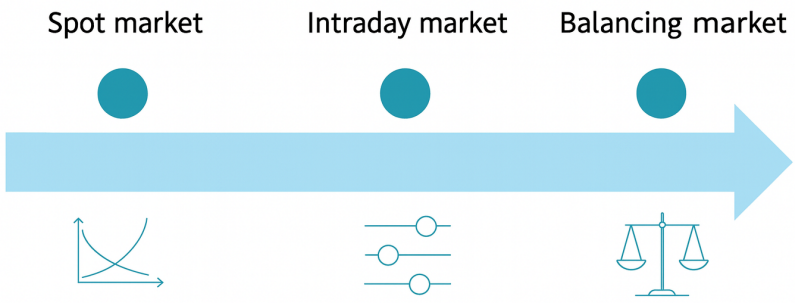


Figure 3: Three main categories in the power market as the delivery hour approaches.

2.1 The Norwegian power market and system

The Norwegian power market was liberalized on the 1st of January 1991, when the Energy Act was officially implemented (Moen & Johansen, 1992). This meant that the supply and demand of electrical energy were subject to the principles of market clearing to increase overall economic efficiency. A clear distinction was thereby established between the competitive supply and demand of electric power through market mechanisms and the monopolistic principle of governing electricity distribu-

tion through the power grid.

The Norwegian power grid can be divided into three main levels: the transmission level, the regional level, and the distribution level. The Norwegian power grid is regulated by The Norwegian Regulathory Authorty (RME), and one of the main responsibilities of RME is to ensure efficient energy distribution management in Norway by facilitating sustainable grid development and improving the utilization of existing power grid. Statnett, the Norwegian Transmission System Operator (TSO), operates and maintains the transmission grid and is responsible for ensuring that the power system remains balanced (Statnett, 2025a). The various reserve markets are described under Section 2.4.

2.2 Day-Ahead market

The Day-Ahead (DA) market is the wholesale market where supply and demand for electricity are determined. Here, producers and consumers submit bids on both volume and price, and then the market is cleared for the coming 24 hours. At 10:00 CET, the available transmission capacities on interconnectors are published, and by 12:00 CET, every market participant must have submitted their bids. Then, the common European coupling process called Single Day-Ahead Coupling uses the Euphemia algorithm to clear the market every hour for the coming 24 hours (Nord Pool, 2023a).

From the 30th of September 2025, the European power system changed its time resolution from hourly to sub-hourly (15 min) (Statnett, 2022). The transition to quarterly time resolution is an important measure to meet the climate targets set by European authorities. More precisely, the measure aims to smooth out structural imbalances in the power system caused by a disaligned production profile compared to the load change.

2.3 Intraday market

To better align supply and demand between the DA market and real-time operations, the intraday (ID) market provides an additional platform for securing the necessary system balance. Once the TSO has calculated the flows based on the results from the DA auction, the available capacities for the ID market are published. The ID

market functions as a bilateral trading platform where participants can trade up to one hour before delivery (Nord Pool, 2023b). The market platform for the ID market is called SIDC (Nord Pool, 2023c).

2.4 Reserve markets

As the delivery period approaches, further market imbalances may arise. Therefore, a range of support services exists to ensure the simultaneous balance between electricity production and consumption, which, in turn, maintains the frequency at 50 Hz in the Norwegian power system. The markets helping maintain this balance are known as the reserve markets and are operated by Statnett. The system imbalances can come in terms of both under- and over-frequency. If the system frequency falls below 50 Hz, balance can be restored by increasing power output to the grid or by reducing power consumption. Conversely, if the frequency rises above 50 Hz, balancing can be achieved by decreasing power output or by increasing the amount of power drawn from the grid.

To achieve a smooth balancing, there is a range of different support services, which can essentially be divided into four main categories: FFR, FCR, aFRR, and mFRR (Statnett, 2023). The set of reserve products all serve different purposes in maintaining a balanced frequency. In Figure 4, it can be observed how an under-frequency is handled through an increase in power from the various ancillary services.

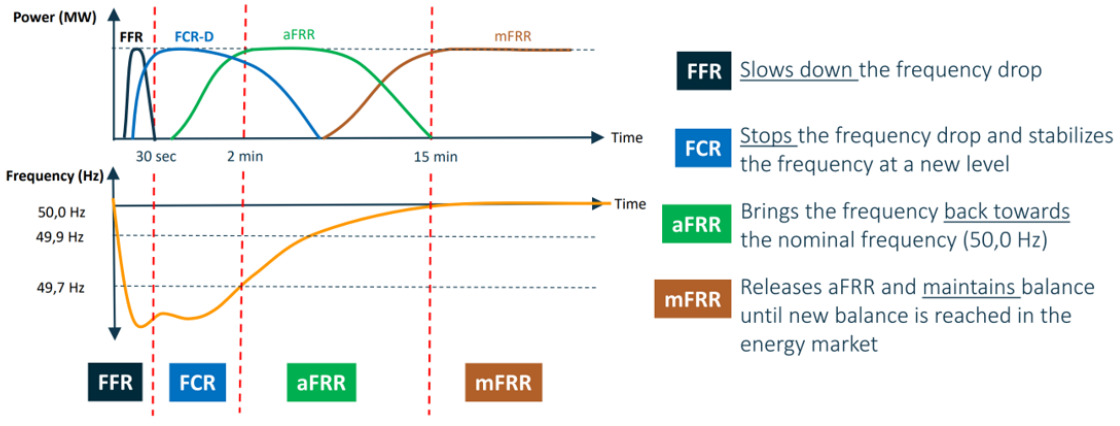


Figure 4: Illustration of various reserve markets activated when an imbalance occurs (Statnett, 2025a).

2.4.1 FFR - Fast Frequency Reserves

FFR is the fastest responding reserve and must be activated within one second. The reserve was introduced quite recently, and exists to ensure that the frequency does not fall below 49 Hz, and is activated once the frequency falls below a certain threshold set between 49.5 and 49.7 Hz. The system operator, i.e., Statnett, is responsible for the procurement of FFR capacity. To limit the possibility of over-dimensioning the procurement of reserve capacity, FFR is acquired in two main products: A seasonal profile with limited capacity on weekends and nighttime (FFR Profil), and a more flexible profile delivered based on request by the system operator and short-term forecasts (FFR Flex). For the seasonal procurement, the minimum bid capacity is 1MW, but for FFR Flex, the minimum bid capacity is 5 MW (Statnett, 2025d).

2.4.2 FCR - Frequency Containment Reserves

FCR is the primary reserve and can be split into two main reserves, FCR-N and FCR-D. FCR-N is activated within the normal frequency band, 49.9 Hz – 50.1 Hz, and is intended to reduce frequency deviations under normal conditions. Thus, providers of FCR-N must be able to supply both upwards and downwards regulation, reflecting the symmetric nature of the product. With larger imbalances related to faults in production units, consumption units, or transmission components, the FCR-D is activated. FCR-D does not have the same symmetric requirement as FCR-N, and for now, downregulation in FCR-D is absent (Statnett, 2025c).

To ensure sufficient capacity from primary frequency reserves, a marketplace has been established to determine the natural intersection between supply and demand of reserve capacity. This market consists of two sub-markets, known as D1 and D2. The minimum bid size in both markets is 0.1 MW, and the maximum response time is 30 seconds. To be eligible for participation in the FCR market, each market provider must be approved by Statnett, act as the balancing responsible for that specific object, and hold an approved prequalification for the specific object (Statnett, 2025c).

The D2 market is executed before the DA market and cleared two days ahead with hourly resolution per bidding zone. Bidders are allowed to submit bids between 00:00 and 17:30 two days ahead of the operational day. Awarded bids are notified by Statnett at 18:30 (Statnett, 2025h).

Further, the D1 market is conducted after the DA market to cover residual requirements, including exchange requests from neighbouring TSOs. The D1 market is cleared daily for the next operational day with hourly resolution per bidding zone. Bidders are allowed to submit bids between 00:00 and 18:00 one day ahead of the operational day. Awarded bids are notified by Statnett by 19:00 this day, which then reveals the price (Statnett, 2025h). The separation into D2 and D1 markets improves predictability because the D2 market runs before the DA auction, while the D1 market runs afterward to cover remaining needs and exchange requests from other TSOs.

2.4.3 aFRR - automatic Frequency Restoration Reserve

Secondary reserves are referred to as aFRR, and they are procured for both up and down regulation. aFRR is activated to restore the frequency to 50.00 Hz. This, in turn, releases the activated primary reserve, making it available for any new faults and imbalances that may occur. The maximum time for full activation is currently set at 5 minutes after the TSO sends a signal. However, Statnett targets a faster response, ideally within 2 minutes or less. The minimum allowed bid size is 1 MW. Bids can be submitted starting from 7 days before the operational day, and must be submitted for each daily auction. Suppliers to this capacity market must submit their bids by 07:30 CET the day before the operational day, but can be modified or withdrawn up until the deadline. The need for aFRR is dimensioned per el-spot area, and Statnett, being the system operator, clears the market before every operating day. (Statnett, 2024a). The resulting market prices are published at 09 a.m. on the operating day.

2.4.4 mFRR - manual Frequency Restoration Reserve

The mFRR, known as the tetriary reserve, releases the activated secondary reserve, aFRR. For mFRR, there exsists two markets: a capacity market and an activation market. In the capacity market, participants are paid for guaranteeing that their reserved capacity can be utilized, with separate markets for both down- and up-regulation. The activation market works by providers submitting bids with prices for regulating production and consumption. The bids are sorted into a common list, and when balancing is needed, the cheapest resources will be activated first (Statnett, 2024b). For mFRR, each bidding zone must maintain its own balance, but can receive

support from neighbouring zones if transmission capacity is available.

The mFRR is the slowest and most energy-intensive reserve. Each bid must be at least 5 MW, have a maximum activation time of 12.5 minutes, and must be procured every 15 minutes. For the mFRR capacity market, bids must be submitted by 07:30 CET the day ahead of the operational day; meanwhile, for the mFRR activation market, bids must be submitted 45 minutes ahead of the operating hour. The bids in the activation market must be submitted with increments of 0.5 EUR/MWh (Statnett, 2024c).

2.4.5 Summarizing the power markets

Section 2 demonstrates the complexity of the power markets that evolve in the pursuit of maintaining a stable frequency in the system. Figure 5 shows the sequential interaction between the markets to fulfill this purpose, while Table 1 summarizes the reserve markets.

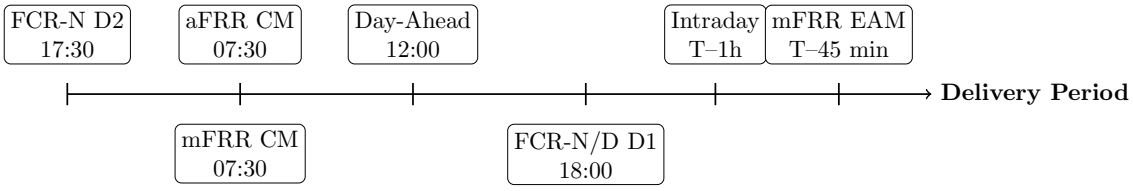


Figure 5: Chronological order of bidding into the power markets.

Table 1: Overview of key informational parameters for the various reserve markets.

	FFR Flex	FFR Seasonal	FCR-N	FCR-D	aFRR CM	aFRR EAM	mFRR CM	mFRR EAM
Response Time	0.7–1.3 s	0.7–1.3 s	$\leq$ 2 min	$\leq$ 2 min	$\leq$ 5 min	$\leq$ 5 min	12.5 min	15 min
Duration	5–30 s	5–30 s	$\geq$ 15 min	$\geq$ 15 min	Bid size[2]	Bid size[2]	Bid size[2]	Bid size[2]
Minimum Volume (MW)	1	5	0.1	0.1	1	1	1	2.5[3]
Implemented (YES/NO)	YES	YES	YES	YES	YES	NO[1]	YES	YES

[1] Planned to go live in 2027/2028

[2] Duration depends on the submitted bid size

[3] Minimum bid of 10 MW constant power for 15 minutes

2.5 Future considerations and research in the power market

The need for electricity and better ancillary services is expected to increase moving further towards 2050 (DNV, 2024). The reserve markets must therefore undergo several structural changes to handle this development. It is still early to determine how these developments and implementations will affect the markets, and they are therefore only discussed briefly in this subsection.

Firstly, the Nordic TSOs are planning to transition to an activation market where the aFRR demand will be calculated per bidding zone. Then, the cheapest bids will be selected for activation through a common optimization, which takes into account transmission constraints between zones. This transition is a part of the Nordic balancing model (NBM), and will take place when Statnett connects to the European Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation, commonly known as PICASSO (Statnett, 2025e). Statnett is expected to join PICASSO in late 2027/early 2028.

Secondly, for the mFRR activation market, the Nordic TSOs will join the Manually Activated Reserve Initiative (MARI) platform. MARI is a European platform for the exchange of mFRR. As with PICASSO, this is also a part of the transition to the Nordic balancing model (Statnett, 2025g). Statnett is expected to join MARI somewhere around the turn of the year 2026 and 2027. Modelling the activation markets is not considered in this project, but rather mentioned as possible future work.

An additional structural change of importance is the transition to a 15-minute time resolution. In the DA market, this change is scheduled for 30th September 2025, while the ID market adopted 15-minute intervals in Norway on 18th March 2025. Since this project relies on hourly data, a finer time resolution could reveal dynamics not captured here. Nevertheless, the long-term implications of this transition remain an interesting topic for future consideration.

Lastly, it is also relevant to consider how reserve market prices have been affected by the transition to flow-based market coupling (FBMC). A preliminary examination of this development may indicate its potential influence on the future power market. On the 29th October 2024, FBMC was introduced in the Nordic region. FBMC optimizes the allocation of grid capacity within and across European countries by accounting for the actual physical power flows in the grid rather than relying solely on predefined bilateral limits. Under FBMC, the allocation of cross-border interconnection capacity

is integrated into the market-clearing process, enabling more efficient use of the grid and reducing price differences between regions (Shinde, 2024). The implementation of FBMC has also affected several reserve capacity markets.

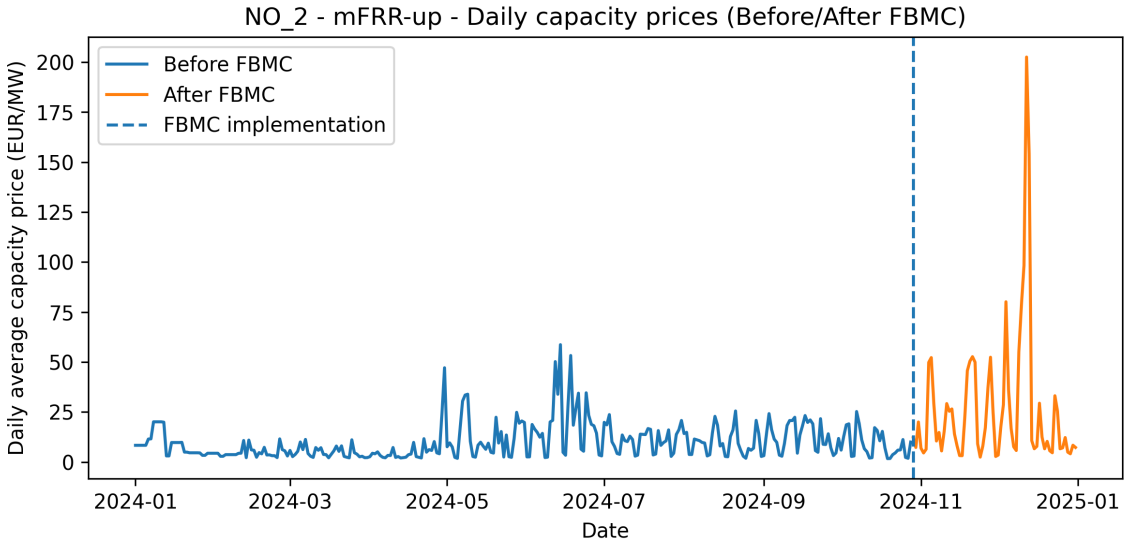


Figure 6: mFRR up prices in NO2 before and after the implementation of FBMC.

Performing a brief, but non-exhaustive, analysis of this development can give some indications on FBMC's influence on future reserve market prices. For instance, Figure 6 illustrates that the mFRR Up market in NO2 experienced an increase in average daily prices after the introduction of FBMC. The same can be observed for NO5 and NO1, but they are excluded for visual clarity. The increase in the daily average prices can be attributed to changes in cross-zonal capacity allocation under FBMC, which may tighten congestion between certain Norwegian price zones and thereby raise reserve market prices. The pronounced spike observed in the plot is associated with the extreme price event on 12th December 2024.

2.6 Demand for balancing services

The project thus far has motivated the need for balancing services, which is in accordance with Statnett's System Development Plan (Statnett, 2025f). It has further outlined how the market design facilitates meeting this need. However, to incentivize

power producers to provide additional balancing services, ownership of flexible resources must be profitable in future market conditions. Identifying whether capacity market prices have increased, alongside attractive price zones, may therefore serve as an indicator of such profitability.

Modelling the future profitability of providing balancing services requires assumptions about the prices that suppliers can expect to receive. Although historical prices may appear reasonable as input data, they are subject to considerable uncertainty. It is therefore important to emphasize that such data may not fully reflect future market conditions. Nevertheless, historical data can be used, provided that it represents market characteristics expected to persist in the long term and that its limitations are clearly acknowledged. The following presents a comparison of capacity prices only used to discuss the appropriate selection of historical data.

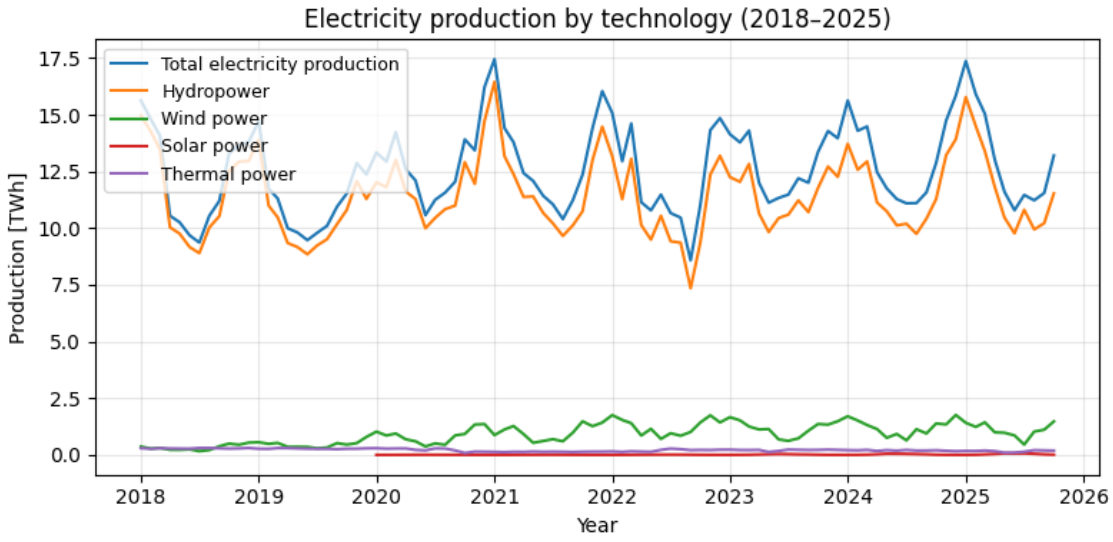


Figure 7: Electricity production mix Norway 2018-2025 (SSB, 2025b).

Firstly, it is necessary to determine which years should be compared. The power mix development presented in Figure 7 shows that most VRES was introduced around 2020. A meaningful comparison of balancing prices should therefore reflect the conditions prior to this development, making 2019 an appropriate reference year.

Furthermore, during 2020 and 2021, two interconnectors were integrated into the Norwegian grid, exposing the power system to structural imbalances arising from the

production mix in continental Europe (Statnett, 2021). In 2022, Russia's invasion of Ukraine triggered a European energy crisis, putting significant upward pressure on power prices. Because of these factors, 2024 has been chosen as the main comparison year to 2019. An exception was made for mFRR and FCR-D due to the unavailability of price data for 2019. Consequently, data from 2022 and 2023, respectively, were used as the basis for comparison.

Analysing the price development

The analysis covers all reserve markets described in Section 2.4, with the exception of FFR and FCR obtained from the D2 market, and includes all Norwegian price zones. To reduce the influence of extreme values, we have excluded observations below the 1st percentile and above the 99th percentile of the price distribution.

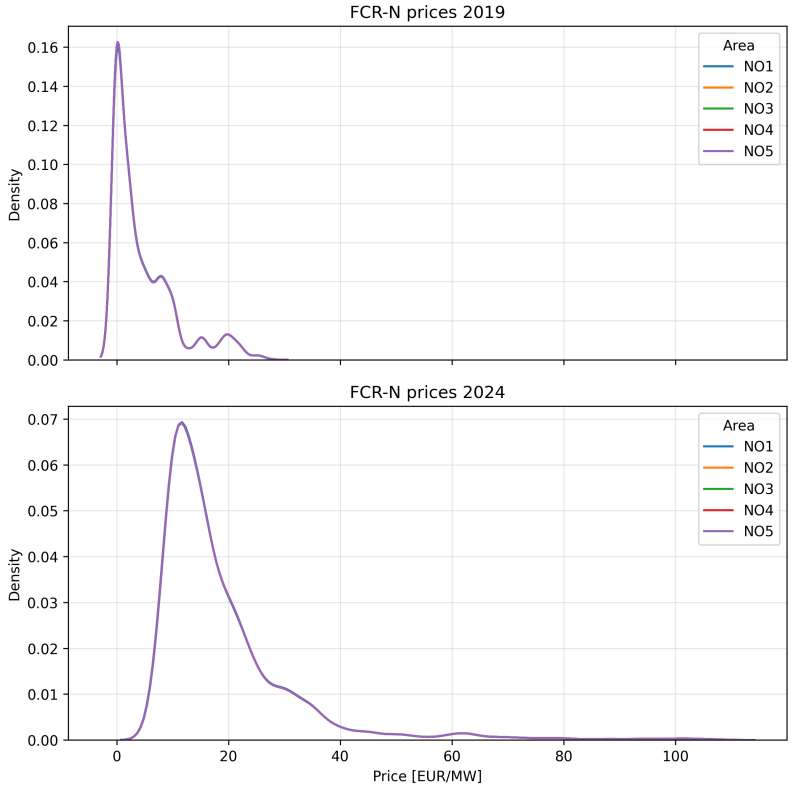


Figure 8: A clear rightward shift in the filtered FCR-N prices from 2019 to 2024.

Figure 8 exemplifies the broader trend of increasing capacity prices from 2019 to 2024. Corresponding plots for the remaining markets, presented in Appendix A,

reveal similar tendencies. This development is further supported by the summary statistics in Table 2. When using 2022 data for mFRR and 2023 data for FCR-D, the trend appears less uniform, highlighting the influence of abnormal price patterns in those years. Based on the 2024 data, the table indicates that FCR-N exhibits the highest average price considering all zones collectively. Further, the table shows that no price zone is distinctly more attractive than the others, while most markets have experienced substantial price increases.

Table 2: Mean reserve capacity prices in $\frac{EUR}{MW}$ for 2019 and 2024.

Year	Market	NO1	NO2	NO3	NO4	NO5
2019 Before	aFRR Up	0.93	1.98	0.74	0.74	1.87
	aFRR Down	0.72	1.84	0.66	0.66	1.70
	FCR-N	4.64	4.64	4.64	4.64	4.64
	mFRR Up*	37.27	40.53	2.39	2.42	36.15
	mFRR Down*	0.67	0.67	0.50	0.50	0.67
	FCR-D**	6.51	8.38	8.28	7.06	7.81
2024 After	aFRR Up	15.71	16.08	18.51	17.64	14.29
	aFRR Down	15.16	12.39	23.08	23.13	8.25
	FCR-N	18.80	18.80	18.80	18.80	18.80
	mFRR Up*	10.02	11.08	3.87	4.16	9.54
	mFRR Down*	5.27	3.96	1.08	2.01	1.64
	FCR-D**	3.07	4.36	5.44	4.65	5.40

* 2022 is used as a comparison year due to missing data from 2019.
** 2023 is used as a comparison year due to missing data from 2019 and 2022.

Input data used for case study

Historical capacity market prices show a nearly uniform increase from 2019 to 2024. Because 2024 better represents current market conditions, these prices are used as input parameters when projecting future revenues from capacity market participation. However, elevated prices may not persist, and increased consumer flexibility could even reduce them (NVE, 2024b). Using historical data can therefore be scaled to reflect alternative future scenarios. Although this is recognised, scenario analysis

lies outside the scope of the project. Consequently, the mid-term hydropower model is solved using 2024 price and volume data.

The model is applied to a case study of a reservoir-based hydropower plant in NO2. Plant-specific data were provided by a confidential power producer in this price zone. The dataset includes maximum and minimum production limits, upper and lower operating water levels, generator elevation, reservoir storage capacity, turbine and generator efficiency curves, and historical inflow time series.

The market prices used as input were retrieved from different platforms. Although full-year aFRR data were used in the analysis found in Appendix A, these data only became available at a late stage of the research process. Consequently, the computational study in Section 9 and the validation in Section 8 rely on simplified representations of the aFRR market. Prices for the aFRR Up and Down markets were collected from NUCS for the 12th day of each month in 2024 and replicated to form annual time series. Similar simplifications were applied to the ID market. Here, volume-weighted average prices were computed using Nord Pool data from the 12th day of each month in 2024, and the resulting hourly price–volume profiles were repeated for the remainder of each month to construct full-year time series. DA prices for the NO2 bidding zone were retrieved from ENTSO-E, while FCR-N and FCR-D prices for 2024 were collected from Statnett. These data were provided on a full-year basis and did not require any further simplifications.

Price data from the mFRR capacity market was not included further in the case study. Among the reserve markets, mFRR is the most energy-intensive, and activated capacity may be dispatched continuously over long periods (Statnett, 2025c). The long and uncertain activation durations of mFRR complicate the requirement to maintain sufficient energy in the reservoir, thereby increasing uncertainty in the estimation of the delta parameters in Section 6.

3 Hydropower

The power output of a reservoir-based hydropower plant depends on the interaction of technical components, physical processes, and economic considerations. Determining an optimal production level while accounting for temporal coupling, non-linear dynamics, and uncertainty is difficult without advanced optimization methods. This section outlines the key elements relevant to the modelling process, beginning with the physical and technical characteristics of hydropower systems and then addressing the associated economic considerations.

3.1 Hydropower from a physical and technical perspective

Over time, the weather-dependent hydrological cycle has shaped the landscape into interconnected systems of oceans, rivers, lakes, and glaciers collectively referred to as water courses (Bonsor, 2001; LVK, n.d.). Hydropower utilizes the topographical differences within these water courses to convert the kinetic energy of water into electrical energy. In Norway, approximately 2700 of the 4000 water courses have been developed for this purpose, leading hydropower to account for about 90% of the national power mix (LVK, n.d. WWF Norge, n.d.). Many of these systems are reservoir-based, where key elements include the reservoir, inflow, penstock, generating unit, and hydraulic couplings. A schematic of such a power plant is shown in Figure 9 to support the following elaboration.

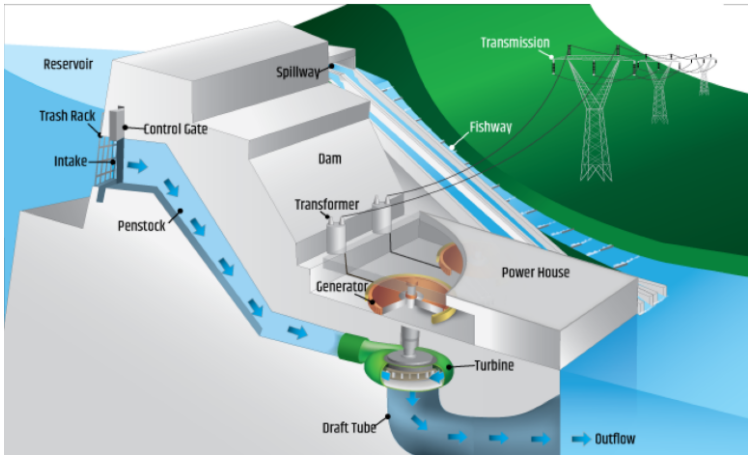


Figure 9: Schematic illustration of a hydropower plant (Kasper, n.d.).

3.1.1 Storage using the reservoir

The reservoir enables water to be stored during periods of high inflow and released during dry periods (LVK, n.d.). It also allows the plant to exploit altitude differences, commonly referred to as the hydraulic head. The water level must remain between the Highest Regulated Water Level (HRWL) and the Lowest Regulated Water Level (LRWL) to satisfy ecological and safety requirements. These limits determine the amount of water that can be stored in a given time step and thereby the available potential energy. In modelling, the reservoir level is typically determined by a balance equation relating inflow, discharge, and storage.

3.1.2 Filling up the reservoir with inflow

Seasonal precipitation patterns and temperature variations largely determine the inflow to the reservoir. The inflow originates from the parts of the water course located at equal or higher elevation. Producers are primarily interested in the utilizable inflow, defined as the portion of water that can be used for production within a given period (NVE, 2024a). Estimating utilizable inflow requires information on snow storage, the unsaturated zone, and drainage sections (NVE, 2015). These factors contribute to determining the feasible production region. Including the avoidance of curtailment, an involuntary reduction in a generator's output caused by grid restrictions, and spillage of excess water influences it even further (Gandhi et al., 2022; IFC, 2015). In modelling, inflow may be represented as a deterministic parameter or as a stochastic variable.

3.1.3 Discharging water through the penstock

To convert potential energy into kinetic energy, water stored in the reservoir is released into a penstock under hydrostatic pressure, directing the flow toward the turbine. Penstocks are typically tunnels or pipelines constructed within the mountains (LVK, n.d.). The cross-sectional dimensions and frictional properties of the penstock determine the instantaneous flow capacity and thereby influence the volume of water that can reach the turbine at any moment. In modelling, the penstock can impose a physical capacity constraint on the rate of discharge.

3.1.4 Converting to electrical energy using the generating unit

The generating unit converts the kinetic energy of flowing water into electrical energy, and consists of a turbine and a generator. Water entering the turbine causes the rotor to spin, inducing an electromagnetic field in the generator and producing electricity (EDP, 2025). The efficiency of the generating unit, defined as the ratio of electrical output to mechanical input, affects the effective available capacity and must be accounted for in the modelling framework.

3.1.5 Using the hydropower production function to determine the generated power

In hydropower modelling, it is essential to represent the relationship between the physical elements described above in a realistic manner. From a flexibility perspective, this is particularly important, as these relationships influence the calculated capacity for participation in reserve markets. However, the underlying physics are far from linear, but instead governed by a complex set of interdependent processes. To capture these dynamics, researchers typically employ a hydropower production function (HPF), which describes how the available water resource is converted into electrical energy through a non-linear relationship (Diniz & Maceira, 2008).

$$P = g \cdot 10^{-3} \cdot \eta_T(q, h) \cdot \eta_G(P) \cdot q \cdot h \quad [\text{MW}] \tag{1}$$

The HPF in (1) expresses the electrical power output as the product of the gravitational acceleration g, the mass density of water absorbed in the factor 10^{-3} for unit conversion, the turbine efficiency η_T, the generator efficiency η_G, the discharge q, and the net head h. The net head depends not only on the geometric altitude difference between the intake and outflow, but also on the reservoir level at each point in time. Because reservoir geometry is often irregular, the relationship between reservoir level and net head is generally non-linear.

Moreover, the turbine efficiency varies with both discharge and head, while the generator efficiency depends on the produced power. These functional dependencies introduce nonlinearity directly into the HPF, making it computationally demanding to include in large-scale optimization models. To maintain tractability, particularly in mid- and long-term scheduling models, it is therefore common to apply simplifying approximations of the HPF.

3.2 Key elements from an economic perspective

The physical characteristics of hydropower provide the basis for understanding the economic decisions faced by producers. Producers must define planning horizons, choose pricing strategies, and identify relevant markets. They must also consider cost drivers, particularly when evaluating investments. The following sections discuss these elements in more detail.

3.2.1 Water value

Section 1 introduced the term water value and explained how it is used to price an otherwise seemingly inexpensive resource by reflecting its alternative use through storage. To understand this concept more thoroughly, it is necessary to consider how seasonal variations affect the reservoir filling level. In spring, electricity prices are typically low and snowmelt generates substantial inflow, causing reservoir levels to rise as production is reduced. During autumn and winter, by contrast, inflow declines while electricity demand and prices increase, leading to higher discharge rates. Although this seasonal pattern is well known, future power prices and hydrological conditions remain uncertain. The core operational challenge is therefore to determine the optimal amount of water to release at each time step to ensure long-term energy security and short-term profitability. By applying statistical analysis and advanced modelling frameworks, producers can estimate the water value, which represents the expected marginal value of having an additional unit of water in the reservoir (A. Helseth, personal communication, April 2024). Thus, the water value reflects the opportunity cost of using one additional unit of water for production rather than storing it for future periods.

3.2.2 Planning horizons

Hydropower planning is relevant to consider at an operational, tactical, and strategic level. Depending on the planning horizon, each level has its own framework, where short-term models deal with the operational concerns, mid-term models are assigned to a more tactical level, and long-term models are dedicated to strategic considerations. Altogether, these models are denoted as Long-, Mid-, and Short-Term Hydro Scheduling optimization models (LMSTHS). It should be noted that although some long-term models may incorporate price-making behaviour, the price-taker as-

sumption is more common across several planning horizons. Under this assumption, electricity prices are treated as exogenous inputs to the model rather than being influenced by the producer's decisions.

Ahlfors and Amelin (2021) describe the different planning horizons and find that the literature is not unanimous in its definitions of these horizons in terms of time. However, the article describes the overarching characteristic of each planning horizon, and a visual representation of the different levels of planning horizons is depicted in Figure 10.

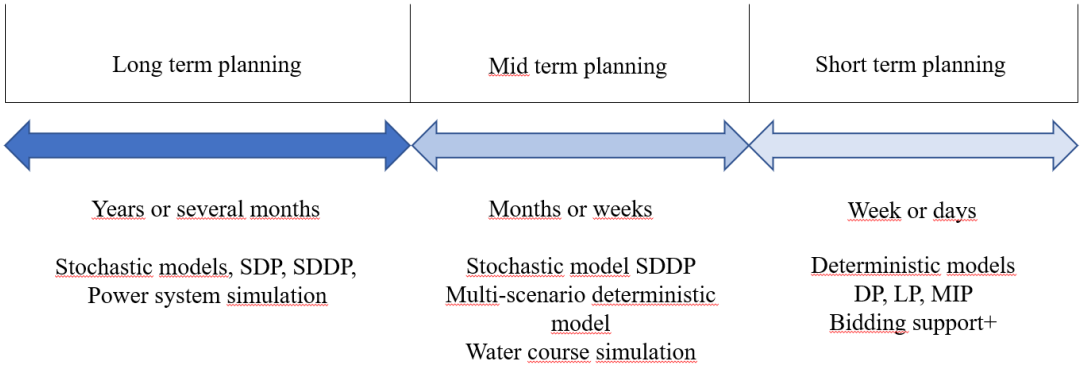


Figure 10: Visualization of long, mid, and short term hydropower scheduling.

Long-term planning

Hydropower planning is inherently stochastic due to uncertainty in future reservoir inflow, which is particularly important to account for in long-term models (Larsen et al., 2015). Many large reservoirs require several years to refill once they have been drawn down, meaning that short-sighted operations can jeopardize future energy security. Consequently, long-term planning and the establishment of water values are essential to ensure that current operational decisions reflect both present market conditions and the long-term availability of water resources.

According to Thaeer Hammid et al. (2020), long-term planning typically covers horizons of one to five years. Its primary purpose is to secure energy availability, and its outputs commonly include water values or weekly discharge plans, which often serve as direct inputs to mid-term models. Because hydropower scheduling involves substantial uncertainty, long-term planning is frequently based on Stochastic Dynamic Programming (SDP) or Stochastic Dual Dynamic Programming (SDDP)

(Warland & Mo, 2016).

Although ensuring energy availability is a central motivation for long-term planning, the mathematical objective of such models can vary from profit to social welfare maximization. Whether both revenues and costs are incorporated in this objective may also vary. When revenues are excluded, long-term models fail to directly capture the potential upside associated with participation in different electricity markets.

Long-term models usually represent hydropower systems at aggregated levels to reduce computational complexity, but this also introduces simplifications that may lead to non-optimal operational decisions (Warland & Mo, 2016). Common simplifications include neglecting detailed hydraulic characteristics or using coarse time resolutions (Ahlfors & Amelin, 2021). In Nordic studies, weekly time steps are common, as illustrated by Helseth et al. (2022), whereas monthly time steps have been observed in a Brazilian system study (Scarcelli et al., 2017). To mitigate the limitations introduced by aggregation, many frameworks employ disaggregation procedures to validate or refine long-term schedules. These procedures help address non-linear and state-dependent constraints that would otherwise need to be simplified or omitted due to problem size and computational burden (Warland & Mo, 2016).

Mid-term planning
This planning horizon typically covers horizons from several weeks to a few months. Compared to long-term planning, the shorter time horizon allows hydropower systems to be represented with a lower degree of aggregation, enabling a more detailed description of hydraulic connections, reservoir dynamics, and operational constraints. However, mid-term planning still involves substantial uncertainty, particularly related to inflow variability and future electricity prices, and therefore relies on many of the same modelling techniques as long-term scheduling. In this context, recent research highlights the role of weekly planning as a promising extension which can offer valuable insights for the development of low-carbon power systems (Ahlfors & Amelin, 2021).

The purpose of mid-term planning is to allocate water resources across seasons in a way that balances operational profitability and long-term water availability. Typical criteria include maintaining adequate reservoir levels and deriving more detailed water values than those produced in long-term models (A. Helseth, personal communication, April 2024). Because the mid-term horizon is closer to real market operations, market participation becomes a central element of the analysis, including participation in day-ahead markets, capacity markets, and financial markets. As

a result, mid-term models are particularly relevant for business case development and investment assessment.

Short-term planning
Lastly, short-term planning concerns the immediate future. As in mid-term planning, the input parameters in short-term models depend on the outcomes of the longer planning horizons. The result is an operational production schedule, typically with an hourly time resolution, specifying the discharge levels for each hydropower plant. Because of the short time horizon, uncertainty plays a less prominent role. Therefore, the modelling focus shifts toward achieving a realistic representation of plant behaviour, including efficiency curves, hydraulic characteristics, and water travel time. Such detailed modelling has become increasingly important as a growing number of hydropower producers participate in both energy and capacity markets (Kong et al., 2020).

Both the objectives and the solution methods vary in short-term hydropower scheduling. Depending on the modelling context, objectives can include revenue maximization, cost minimization, or minimization of spilled energy (Kong et al., 2020). Although short-term models are commonly solved deterministically, accounting for uncertainty can improve their performance (Belsnes et al., 2016). Heuristic methods are also used in the literature to address the STHS problem, as they are able to provide good solutions within short computation times (Kong et al., 2020).

3.2.3 Market participation

Reservoir-based hydropower, with its significant storage capacity, is capable of providing considerable operational flexibility. In addition to the day-ahead (DA) and intraday (ID) markets, hydropower producers can participate in several reserve markets, as highlighted in Section 2. According to the Nordic TSOs (2019), a large share of Norwegian hydropower capacity qualifies for participation in the FCR market. Furthermore, because hydropower can deliver the required response times, it is also eligible for the aFRR and mFRR markets, which have longer activation times compared to the FCR products.

3.2.4 Hydropower investment analysis

Many of the large hydropower plants in Norway were built between the 1950s and 1980s, and many are now approaching or exceeding their technical lifetimes. Because political regulations limit further development of many of the remaining watercourses, reinvestments have become the primary strategy for hydropower producers (WWF Norge, n.d.). This strategy is further reinforced by the growing need for system flexibility, which increases the importance of upgrading and reconstructing existing hydropower plants (Statkraft, n.d.). Such reinvestment measures fall into the two categories, rearmament and expansion (NVE, 2025b).

Rearmament
Rearmament focuses on improving existing parts of the power plant while keeping the discharge and head at the same level. This can include replacing the impeller, replacing the generator, or reducing the friction loss in the penstock. These measures typically fall under the same concession and are therefore exempt from new licensing procedures (NVE, 2020).

Expansions
Expansion measures can enable higher discharge capacity and/or a greater head. This may include enlarging the cross-section of the penstock, installing pumped-hydro storage, which effectively reduces the LRWL, or extending tunnels to increase the available head. "Driva 1 TWh" is a concrete example of a planned capacity expansion in a Norwegian hydropower plant (TrønderEnergi, 2025). Unlike rearmament, these measures require new licensing procedures (NVE, 2020).

Main cost-drivers
There is limited publicly available information about the costs associated with rearmament and expansion that can be directly applied in an analysis. According to a report published by the International Renewable Energy Agency (IRENA) (2012), the estimated cost ranges between 500 and 1 000 USD/kW of increased capacity. When adjusted for inflation and converted using an exchange rate of 0.85 EUR/USD, this corresponds to approximately 625 000–1 250 000 EUR/MW. This is, however, a very broad range, as the cost is heavily dependent on which investment is considered. According to IRENA (2012), the cost structure of a hydropower refurbishment or expansion project is largely governed by two main drivers: civil works and electromechanical equipment.

- **Civil works costs** includes all structural and construction-related activities such as dam or reservoir construction, tunnel construction, powerhouse modifications, access road improvements, grid-connection works, as well as engineering, procurement, and construction services. For upgrade projects, relevant measures may include redesign of water intakes, refurbishment or enlargement of tunnels, and structural interventions on the dam. Because civil works often involve extensive excavation, concrete work, and site-specific engineering, they frequently constitute the most significant share of total investment costs.
- **Electro-mechanical equipment** includes the costs associated with replacing or upgrading turbines, generators, transformers, cabling, and control and protection systems. Turbine and generator replacements are particularly relevant for capacity upgrades, as modern designs can provide improved efficiency, higher output, and enhanced operational flexibility without requiring major modifications to the existing waterway infrastructure.

NPV - Net Present Value Method

To quantify the value of a reinvestment, calculating the net present value (NPV) is highly relevant. The NPV method aims to measure whether a project will be profitable by comparing its future cash flows to its cost and will be applied to the case study later in Section 9. The formula for NPV can be summarized as follows (Investopedia, 2025):

$$NPV = \sum_{i=1}^{N} \frac{CF_i}{(1+r)^i} - I_0 \tag{2}$$

I_0 reflects the initial investment cost of the upgrade. CF_i is the net cash flow, i.e., the additional periodic income from the upgrade corrected for the additional operational costs. This cash flow is then discounted at the discount rate r, which reflects the capital cost of the reinvestment. These cash flows are computed for the entire planning period and discounted back to the present value. If the net present value is positive, the project should be considered for implementation.

4 Related literature review

The literature on hydropower planning covers a variety of topics. This section presents a related literature review of topics relevant to this report. While Section 3 highlighted aspects related to the planning horizons, the review will further focus on the dimensions of hydraulic modelling, stochasticity, market participation, and solution methods, presented through the 5'Cs framework (A. Gullhav, personal communication, October 2025). Each dimension will cite, compare, contrast, and criticize the content presented in the literature and connect it to the work presented in this project.

4.1 Searching strategy to find relevant literature

A two-step searching strategy was applied to find relevant literature efficiently. First, we chose a set of particular searching words in databases such as *Google Scholar, Science Direct, and Scopus*. Search words that have been used are *HYDROPOWER SCHEDULING, CAPACITY MARKETS, UNCERTAINTY, PLANNING HORIZONS AND PROFIT MAXIMIZING, MULTIMARKET OPTIMIZATION*. Additionally, Elicit has also been used for finding relevant articles. Elicit is an AI model that utilizes a large language model to retrieve relevant articles. The search in this model was: *Multimarket Optimization of hydropower across the Norwegian Power markets*. When picking relevant articles or papers, we have asked ourselves quality check questions:

- Does the title and abstract fit our purpose?
- Is the document published in a recognized journal that practices peer review?
- Does it contain a broad literature list with other recognized articles?
- Does it contain a clear literature review on the topic it is writing about?

While the first two questions are relevant to all the selected articles, only one of the latter two had to be fulfilled. Once approved accordingly, we started to read qualified papers. The second step was to identify new potential papers based on the referencing in the initial paper using similar control questions. The method that has been used is an adaptation of the backwards snowballing method, which is a generalized method for performing literature studies (Wohlin, 2014).

4.2 Stochacitic

Capturing the stochasticity associated with hydropower planning is highly relevant in an operations research setting. While deterministic models often provide useful insights to decision-making, Belsnes et al. (2016) showcased in their study that adding stochasticity enhances profit margins by allocating production to expensive hours and reducing the risk of spillage. This was demonstrated in a case study by evaluating the strategies obtained from the stochastic and the deterministic models over a new set of scenarios. Consequently, the stochastic model provided more robust solution strategies to the decision-maker while improving the degree of realism.

An important element of uncertainty in hydropower planning is inflow. This is frequently recognized in literature review across all types of planning horizons (Abgottspon & Andersson, 2013; Belsnes et al., 2016; Helseth et al., 2016, 2017). Inflow to a given reservoir is determined to a large extent by the temperature conditions in the area where the hydropower plant is located. Hence, temperature can also be treated as a stochastic variable in a hydropower planning model, as seen in the SOVN model (Warland & Mo, 2016). Temperature is also closely linked to electricity consumption patterns, influencing the demand curve used to determine the market-clearing price (Idsø et al., 2024). This relationship is particularly relevant in long-term models, where prices are derived through a market-clearing process, as seen in the EMPS model (SINTEF, 2025).

Building on this, it provides a rationale for modelling prices as stochastic variables in short- to mid-term hydropower optimization models. Schäffer et al. (2022) implement stochastic market prices in their study, in addition to a generic stochastic environmental variable, which implicitly represents temperature by indicating whether a low inflow period has ended. Aasgård (2020) and Faria and Fleten (2011), on the other hand, do not include any explicit environmental variables and instead focus on market prices. Various configurations of modelling stochastic market prices have also been observed in several of the papers mentioned above. Ideally, a model should take all prices across the existing markets into account, as they are subject to uncertainty (Belsnes et al., 2016). However, treating all prices as independent stochastic variables would possibly yield the curse of dimensionality when generating scenarios for the problem.

Hence, the literature appears to converge on the importance of incorporating stochasticity into models across different planning horizons, as this improves the realism of the results. The most prominent stochastic variables to include are prices and inflows.

Uncertain prices are particularly relevant in models where actors operate as price takers, while uncertain inflow is important to consider irrespective of how the price is determined. Regardless, researchers often limit the number of stochastic variables to avoid the curse of dimensionality.

4.3 Hydraulic modelling

Section 3.1.5 showed the non-linear relationship between generated power and net head. To achieve sufficient precision of a hydropower system, accurate hydraulic modelling is often essential, as pinpointed in a meeting with Professor Arild Helseth at the NTNU Department of Electric Energy (A. Helseth, personal commmunication, 07th of November 2025). When incorporating the HPF into a model, it becomes crucial to select a method that both captures the underlying physical relationships and aligns with the problem requirements. This is often a trade-off between computational complexity and the desired level of model detail.

Several techniques for modelling the HPF exist, which depend on the time horizon, temporal resolution, system size, the available data, and the goals of the optimization model (Kong et al., 2020). The study points out piecewise linear approximation as a common technique to account for the non-linear HPF. The breakpoints, defined as pairs of water discharge and power output, are typically specified in advance and represented either as a two-dimensional table for fixed-head conditions or as a three-dimensional matrix when head variations are considered (Kong et al., 2020). A closely related approach described in Chazarra et al. (2016) is discretizing a family of non-linear curves and using binary auxiliary variables to indicate which section of the efficiency curve is activated for a given time step. As such, this approach can be characterized as a mixed integer linear program (MILP). The drawback of a MILP is that it often can become computationally burdensome when the size of the problem increases (Kong et al., 2020). It is important to consider this in relation to the purpose of the hydropower planning model.

In contrast to the MILP approach, Gjelsvik and Haugstad (2005) and Belsnes et al. (2016) present a less computationally demanding approach to the HPF modelling. Gjelsvik and Haugstad (2005) present ProdRisk, a mid-term hydropower scheduling model developed by Sintef Energy Research. It ignores the variable head in the HPF and introduces correction coefficients in the form of head sensitivities instead. This is done through two separate simulations, in which the last simulation aims to correct the initial simulation through the inclusion of head sensitivities. Similarly,

Belsnes et al. (2016) use an iterative method called successive linear programming, where the efficiency related to the net head is updated based on the efficiency used in the previous time step. To avoid flip-flopping solutions and ensure convergence, the efficiencies are linearized using a first-order approximation of the coupling between efficiency and a marginal increase in reservoir level. In addition, a penalty is added to the objective function for deviating from the previous efficiencies in the iterative process. The study concludes that implementing a head-of-water optimization improves the model solution, especially when the reservoir is half-full due to its flexibility.

In the continuation of this, it is recognized that modelling hydraulic behavior can be decisive for model performance, but some approaches might lead to increased computational complexity. Thus, this trade-off needs to be tailored to fit the purpose of the model.

4.4 Multimarket dynamics and participation

Norwegian hydropower producers operate across energy and reserve markets, leveraging the storage capability to maximize revenues (SINTEF Energi, 2025). Modelling multiple markets can therefore provide more accurate income estimates and a more realistic representation of hydropower's true value than single-market approaches. Consequently, reviewing how the literature addresses multi-market participation across modelling frameworks is essential to capturing the full value-creation potential that this work aims to explore.

It is evident that the literature provides studies with different configurations of market participation. Aasgård (2020) investigates a multi-market bidding problem considering the DA-market and mFRR-market using a price-taking approach, without the inclusion of reserve activation. Her model co-optimizes the revenue gained from participating in both markets, and a case study from NO2 shows positive gains from such coordinated bidding. Chazarra et al. (2016) report similar results when modelling coordinated bidding in DA-markets and the secondary reserve markets (aFRR) in a Spanish 7-day case study, as this yields significantly higher income compared to DA-participation only. Although the number of connected reservoirs and power plants may vary between the studies, they both show the benefits of multi-market optimization. A potential limitation with the latter model, however, is that it assumes that an expected fixed percentage of the contracted reserves is always activated. While this exogenous parameter is determined through statistical analysis, this is a simplified behaviour of the market, potentially yielding too optimistic results. On the

contrary, the model mitigates risk in terms of water shortage and spillage, thereby reducing the overall risk associated with the strategy suggested by the model.

Building on the modelling of activation presented in Chazarra et al. (2016), Farahmand et al. (2020) emphasizes the value of detailed reserve market activation modelling, taking the perspective of the system operator. In particular, the study demonstrates that reserve provision decisions that ignore activation lead to systematically underestimated costs. To a hydropower producer, this manifests as lost water value, adjusted reservoir management, slightly reduced future flexibility, and increased ramping. The latter can lead to additional wear and tear on the hydropower unit.

Similar to several of the above-mentioned studies, Faria and Fleten (2011) also present a co-optimization model across the DA market and the Elbas market (Intraday), from a price-taking perspective. The model captures both block and hourly bids in a 24-hour schedule in a rolling horizon fashion over multiple consecutive days. In contrast to Aasgård (2020), this model is implemented on a case study with a hydropower plant located in NO1, and the results indicate that including ID-participation yields negligible increased profit margins. While these studies seem to disagree on the benefit of modelling multi-market participation, it is important to emphasize that they model different markets and bidding zones. Thus, the results might not be directly comparable, but they highlight interesting aspects of multi-market interaction.

While the previous studies have considered the value of adding one extra market into the optimization model, Ahmadi et al. (2015) choose a broader approach to multi-market participation by including the DA, the bilateral long-term contracts, and spinning and non-spinning reserve markets. The focus in their study, however, is multi-objective optimization where minimizing environmental impact from thermal power plants serves as the second objective. Thus, the study lacks a discussion on the added value of modelling several power markets. The direction of the balancing market is also a dimension that is omitted in this study.

Long-term forward contracts and price-making hydropower producers fall outside of the scope of this project. However, in the literature review, it seems worth mentioning the work of Abgottspon and Andersson (2013). They showed that adding forward contracts as an ingredient in the operation of a price-making hydropower plant, as well as pumped storage, contributes to a 20% increase in profit. While the model tends to be realistic and tractable, the results might be coloured by lin-

ear dependencies between commodities and market prices, and the assumption that forward contract prices are known.

Connecting these articles reveals that the additional economic value of a joint market formulation across eligible markets for a hydropower producer is largely absent from the literature, as also noted by Belsnes (2016). Such a formulation could specifically integrate the various capacity markets with participation in the DA and ID markets. It would be particularly interesting to examine whether the synergies from co-optimization yield more accurate estimates of the potential operational revenues for a hydropower plant. In our work, however, it is important to note that omitting activation markets may lead to underestimation of operating costs.

4.5 Relevant solution methods

Within hydropower planning, a range of solution methods is used to handle short- and mid-term operational problems, and long-term strategic decision-making. The choice of method shapes how flexibility, uncertainty, and the value creation of the hydro resources are represented. Reviewing these methods in the literature is therefore important for understanding how different approaches can be utilized, and for positioning our own work.

SDP/SDDP

A common solution method used across all hydropower planning horizons is Stochastic (Dual) Dynamic Programming (SDP/SDDP) as seen in various studies (Helseth et al., 2016, 2017, 2022; Pérez-Díaz et al., 2020). Helseth et al. (2016) use the SDDP framework for profit maximization of a hydropower producer selling electricity and capacity in separate markets. The problem is decomposed into weekly subproblems for a given sample of scenarios, and solved through a forward iteration where information is passed from one state to another. Then, at the end of the planning horizon, a backward iteration is performed, and cuts are added to the problem based on sensitivities from the state variables on the objective function. Similarly, Helseth et al. (2017), a SDP-model is used to assess the operational profitability of a hydropower producer participating in the energy and capacity market. The complete planning period is one year, but the problem is decomposed into weekly MILP problems. The SDP decomposition allows for the representation of non-linear and non-convex relationships, and for the given week, the realizations of the stochastic variables are known.

Péres-Diaz et al. (2020) also use the SDP-framework in their mid-term optimization model, which has a scheduling horizon of one year. However, they decompose the problem into daily decision problems as mixed-integer quadratic programming problems with known exogenous stochastic state variables. These variables were the daily average energy market price, the daily inflow, and the daily residual demand curve in the reserve market. While both SDDP and SDP are common solution methods for stochastic operational decision problems, SDDP might be superior to SDP in terms of computational cost. The SDDP approach increases linearly with the number of scenarios, while the SDP increases exponentially.

Two-stage Benders decomposition
A conceptually related solution method to SDP/SDDP is the two-stage Benders Decomposition method, identified in numerous studies (Jacobson et al., 2023; Sifuentes & Vargas, 2007; Warland & Mo, 2016). Warland and Mo (2016) show that the Benders decomposition is a suitable solution method to long-term hydro-thermal planning problems. Using the stochastic optimization model for Scandinavia with individual water values and grid restrictions, denoted as SOVN-model, a sequence of problems referred to as scenario fan problems (SFP) is simulated for each time step in the planning period. Each SFP problem is a two-stage stochastic LP problem solved by using Benders decomposition. The first stage represents a given week of the simulation (weather scenario and week), and the second stage is represented by independent scenarios, where the decisions from the first stage problem are taken as input and used to solve the rest of the planning period for each scenario. Their analysis shows promising results, as it yields a better utilization of hydro resources compared to EMPS. This is also a long-term planning model, but it is based on aggregated hydro description and heuristics.

In Jacobson et al. (2023), a Benders Decomposition technique for a large-scale energy system planning problem is presented. The complete problem is formulated as a MILP model. Further, it is then decomposed into a master problem, which handles investment decisions, and operational subproblems, handling detailed operational constraints. A key innovation in this work is the decomposition of the operational problem by weeks, despite the planning horizon covering the full 52-week year. This is achieved by introducing budgeting variables that impose weekly limits, thereby allowing the weekly subproblems to be solved independently and in parallel, Jacobson et al. (2023).

Sifuentes and Vargas (2007) shows that the Benders decomposition is also applicable when solving a short-term hydro scheduling problem (STHS) with a full AC power

flow model. The master problem is a mixed-integer programming problem that minimizes fixed costs associated with hydropower generation and thermal units. Whereas in the subproblem, the Optimal Power Flow (OPF) problem is solved. Then, the Benders cuts are applied to the master problem with respect to the sensitivities of the dual variables obtained from the OPF solution. These studies show that, irrespective of the planning horizon, Benders decomposition could be a relevant solution method.

Stochastic Successive Linear Programming
Building on the short-term solution methods presented above, Belsnes et al., (2016) propose a stochastic successive linear programming approach for the STHS problem, explicitly modelling uncertainty via scenario trees. They highlight a limitation of scenario-tree-based multistage stochastic programming, namely the decay of uncertainty as one moves deeper into the tree. The effective uncertainty is reduced, which biases the solution towards determinism. To mitigate this, they discuss the use of rolling horizon schemes, recalculating scenario trees as new information becomes available. They also note that scenario-tree generation and fine temporal resolution raise computational tractability issues when the horizon is very long, or the discretisation is fine, akin to the limitations of SDP. However, the SSLP does not exploit structures in hydropower planning, which avoids the benefit of decomposing the large hydropower planning problem. Thus, this method is primarily suited for STHS problems.

Multi Objective Programming
While the presented solution methods thus far have focused on single objective optimization, Ahmadi et al. (2015) choose a different approach to the hydropower scheduling problem. They present a deterministic multi-objective optimization (MOO) approach that simultaneously considers emission minimization and profit maximization. In their work, the Normal Boundary Intersection (NBI) method and Weighted-Sum (WS) method are applied to a MILP problem to generate non-inferior solutions and construct the Pareto-optimal frontier. Obtaining this frontier provides stakeholders with valuable insights into the trade-off between profit and emissions. While considering emissions is beyond the intended scope of our study, MOO represents an important direction for future research, particularly when integrating sustainable energy sources such as wind and solar (Ahlfors & Amelin, 2021).

The presented solution methods in this subsection show that there are several approaches to the hydropower planning problem, depending on the inclusion of stochasticity, nonlinearities, multistage, planning horizon, and objectives. SDDP is a prom-

ising solution method if all these characteristics are under consideration, while SDP is suitable as long as the planning horizon is short enough to avoid intractable scenario trees. This is also a shortcoming of the SSLP. In contrast, the Benders' decomposition method avoids the planning horizon issue. Simultaneously, the method discussed here is limited to two-stage problems. However, this limitation can be addressed by using a nested Benders approach. Nested Benders essentially extends the classical Benders decomposition method to a scenario tree structure, where the tree can be viewed as a set of nested two-stage problems (Murphy, 2013). If several conflicting objectives are under consideration, MOO techniques such as the NBI and WS can be used to construct Pareto frontiers. In total, these methods provide guidance in how various problems can be solved in a holistic and more realistic manner.

4.6 Our contribution to the literature

An overview of the discussed literature review is provided through Table 3 where the stochasticity, hydraulic modelling, market participation and solution methods are compared. Further, it displays the contribution this report has to the existing literature.

To the author's best knowledge, there is a research gap concerning multi-market participation for hydropower producers that simultaneously considers all eligible capacity markets without having to explicitly model activation. Since activation is excluded in this work, model complexity is reduced by omitting the mFRR market. As most power exchange occurs in the DA market, the reviewed literature shows that its inclusion is essential in hydropower planning models. Furthermore, while several studies examine participation in individual capacity markets, none incorporate the DA, ID, FCR, and aFRR markets jointly. Ahmadi (2015) modelled a detailed market participation framework, but within a multiobjective deterministic context and without distinguishing upward and downward regulation.

Thus, Section 4.4 reveals an opportunity to model a more exhaustive market participation, and study how this affects water values, operational patterns, and potential revenue streams discussed from a hydropower capacity expansion perspective. This is pursued in the case study, which uses a mid-term planning horizon with hourly time resolution.

The remainder of the literature review points out important aspects and solution techniques to achieve realistic results for the hydropower planning problem. In our

modelling, we recognize the importance of representing hydraulic behavior in reservoirs as found in Section 4.3. Further, the authors realize that this is especially important in the consideration of capacity market participation, as the level of accuracy might affect the actual available capacity for participation. While the purpose of this project can be achieved by solving a deterministic model, Section 4.2 showed how pervasive stochastic modelling is in hydropower modelling to achieve accurate results. Section 4.5 presented several possible solution methods if stochasticity is to be added to the model.

Table 3: Review of research papers, highlighting how this report contributes to the existing literature.

Article/ Reference	Planning horizon	Hydraulic modelling	Stochastic	Market participation	Solution method
(Aasgård, 2020)	STM	[-]	Prices and Volumes	DA and mFRR	DE
(Faria & Fleten, 2011)	STM	Constant head	DA and ID prices	DA and ID	DE
(Helseth et al., 2016)	MTM	Constant head	Inflow and Prices	DA and CM	SDDP
(Helseth et al., 2017)	STM/MTM	Variable head	Inflow and Prices	DA	SDDP
(Belsnes et al., 2016)	STM	Variable head	Inflow and Prices	DA	SSLP
(Abgottspon & Andersson, 2013)	STM/MTM	[-]	Inflow and Prices	DA and FM	SDP
(Schäffer et al., 2022)	MTM	Constant head	Inflow, Prices and Environmental state	DA	SDP
(Warland & Mo, 2016)	LTM	[-]	Inflow, Prices and Temperature	EM	BD
(Ahmadi et al., 2015)	STMO-M	Variable head	[-]	DA, RM and FM	NBI
(Chazarra et al., 2016)	STM	Variable head	Inflow and Market prices	DA and aFRR	SDP
(Jacobson et al., 2023)	MTM	[-]	[-]	EM	BD
(Helseth et al., 2022)	STM	Constant head	Inflow	System level	SDDP
(Pérez-Díaz et al., 2020)	MTM	Variable head	Inflow, market prices and residual demand	EM and RM	SDP
(Sifuentes & Vargas, 2007)	STM	[-]	[-]	[-]	BD
Our contribution	MTM	Constant Head	[-]	DA, ID, FCR and aFRR	LP

Abbreviations: STM = Short-term model, MTM = Mid-term model, LTM = Long-term model, LP = Linear programming, BD = Benders decomposition, DE = Deterministic equivalent, SDP = Stochastic dynamic programming, SDDP = Stochastic dual dynamic programming, SSLP = Stochastic Successive Linear Programming, NBI = Normal Boundary Intersection, STMO-M = Short-term multi-objective model, DA = Day-ahead, ID = Intraday, CM = Capacity markets, RM = Reserve Markets, EM = Energy Markets

5 Problem description

This section presents a detailed description of the multi-market participation problem for a single hydro power plant, with emphasis on key limitations and decision-making processes involved. Assumptions that have been made when developing the model and their implications for the results are highlighted in Table 5.

The operator of a reservoir-based hydropower plant must coordinate production decisions across operational, tactical, and strategic horizons. The present discussion focuses on the mid-term operational setting, in which the plant is scheduled on an hourly basis across a full year. In this context, the operator receives inputs in the form of expected water inflow to the reservoir and market prices in the day-ahead (DA), intraday (ID), and capacity markets (aFRR and FCR). These price signals and hydrological conditions evolve over the planning horizon and constitute the informational basis for subsequent operational decisions.

In reality, prices and inflows are uncertain. However, assuming a fairly certain estimate exists, the operator must determine the hourly allocation of energy and capacity across the available electricity markets. This entails selecting the production level from the reservoir and distributing available power capacity between energy and capacity markets. The central decision variables, therefore, include the amount of energy produced and sold in the DA and ID markets, as well as the upward and downward committed capacities in the aFRR and FCR markets. It is assumed that the operator can submit bids for every hour simultaneously and does not have to consider the chronological sequence of the market participation as described in Section 2.

These decisions are governed by a set of physical, hydrological, and market-related constraints. First, generation is bounded between minimum and maximum plant capacity, and the efficiency of the turbine depends on the level of production. Second, the reservoir state must remain within prescribed upper and lower limits for operational, environmental, and safety reasons. Water balance dynamics link decisions intertemporally. While water release for electricity generation reduces the reservoir level, inflow replenishes it, meaning that the level at any hour directly influences the feasible operating region in subsequent hours. Third, participation in capacity markets is contingent upon the scheduled production level. The ability to provide upward reserve capacity is constrained by the margin between the plant's current production and its maximum generation capacity. On the contrary, downward reserve capacity is bounded by the difference between current production and the minimum

stable operating level. In both cases, reserve capacity may be activated. While the activation dynamics are kept outside the model, the possibility of activation can be reflected by introducing additional margins to the minimum and maximum stable operating levels of the reservoir to ensure sufficient energy security. Additionally, the size of the plant limits the share of the reserve market that the plant can capture. Lastly, as the plant operates as a price taker, there is a tradeable volume limitation in the ID market based on a set percentage.

Within this framework, the operator's objective is to maximize total revenue over the planning period by optimally allocating energy and reserve capacity across the markets while respecting physical and regulatory constraints. The result identifies the optimal production schedule and market commitments for each hour. This specifies the reservoir state, discharge, and energy and capacity allocations that jointly represent the economically best operating strategy under the given conditions.

6 Mathematical model

The model is designed as a deterministic linear programming model, from the perspective of a price-taking hydropower producer. The model aims to maximize operational revenue across energy and capacity markets. Since the model is given perfect foresight of market prices, the results obtained from running it represent a demonstrative upper bound on the potential revenues that a power plant can theoretically achieve. In addition, the model introduces PQ-curves to more accurately model the relationship between active power delivered to the grid and the water discharged from the reservoir.

6.1 Model assumptions and simplifications

The assumptions used in the model regard market dynamics and power plant specifics. These assumptions and consequences are presented in Table 5.

Table 5: The underlying assumptions and their consequences for the mathematical formulation.

Assumption	**Consequence/comment**
The hydropower production function depends solely on reservoir discharge, implying only minor variations in head over the planning horizon. This assumption was confirmed as reasonable following discussions with Professor Arild Helseth at NTNU (A. Helseth, personal communication, 7th November 2025).	This simplification may cause slight deviations from the actual PQ-curve, but this is not expected to be significant given the characteristics of the plant.

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Assumption	**Consequence/comment**
The state of the reservoir can be described using the reservoir state in the previous time step plus inflow minus the discharge.	Since inflow is represented using discrete time steps, the inflow at time t can simultaneously be used for market trading. The model does not explicitly capture the time coupling between water inflow and production.
Only capacity markets are modeled, while energy activation markets are neglected	As a result, the model neglects potential income from delivering energy through activation. However, due short activation time in the aFRR and FCR markets, these revenues constitute a rather small contribution
The operator is assumed to have perfect information regarding day ahead and intraday prices, such that the actual production delivered to the market equals the day-ahead obligations minus the net intraday position. The operator is also assumed to have perfect foresight regarding capacity market prices.	Assuming perfect information leads to an optimistic upper bound on revenues, and a potential overestimation of the true revenue streams. Additionally, it allows the model to explicitly exploit the possibility of arbitrage between DA and ID markets.
The power producer operates as a price taker.	This implies that the producer cannot influence the merit order curve, and the model does not attempt to capture strategic bidding behaviour.

PQ-curve modelling

For the given power plant, there exists a relationship between the amount of water drained from the reservoir and the power output delivered to the grid. This relationship can be described using a power discharge curve, denoted as a PQ-curve. Using the data presented in Appendix B and Appendix C, combined with linear interpol-

ation, points describing the relationship between discharge from the reservoir [m^3/s] and the power [MWh/h] delivered to the grid were obtained.

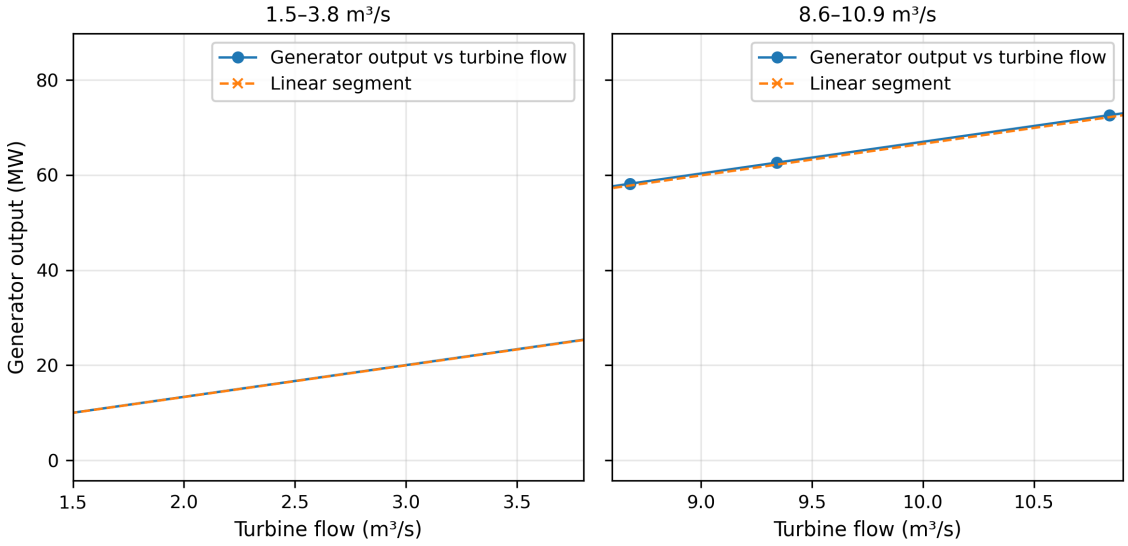


Figure 11: The production relationship for two separate sections of the PQ-Curve.

The observed blue PQ-curve in Figure 11 is compared to a simple orange linear PQ-curve approximation. This is plotted for two separate segments of the PQ-curve. The plot suggests that the PQ-curve appears to exhibit a linear to weakly convex shape for low discharge values (left plot), and a concave relationship for higher levels of water discharge (right plot). To incorporate these characteristics into the optimization model, the relationship between reservoir discharge and power production must be represented explicitly, either through the introduction of binary variables or through a simplified relationship.

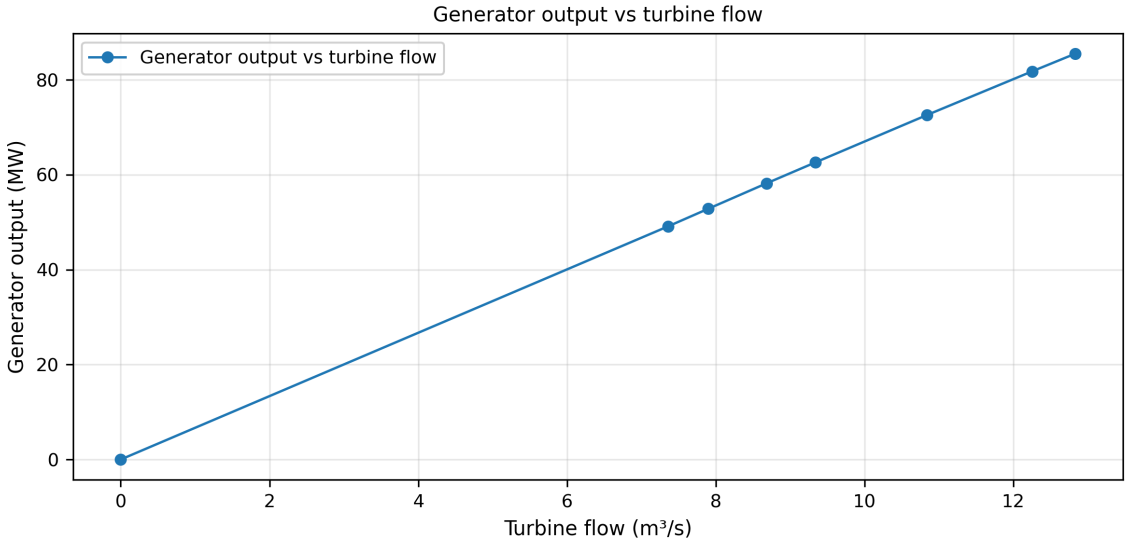


Figure 12: The simplified PQ-Curve used in the mathematical model.

To avoid the use of binary variables, which would increase computational time and pose challenges for future model expansions, it was decided to approximate the weakly convex part of the production curve using a simplified linear relationship. This approximation is estimated to give small deviations from the true PQ-curve. For the concave part of the curve, a set of piecewise linear inequalities was constructed. These inequalities provide a conservative representation of the true system performance while maintaining linearity within the model. The PQ-curve used for modelling in this work is presented in Figure 12.

Sets

Table 6: Defining the sets used in the model.

Set	Description	Elements
$\mathcal{T}$	Set of time periods with hourly step length	$t \in \mathcal{T}$
$\mathcal{E}$	Set of energy markets available for participation consisting of DA, ID Sell and ID Buy	$e \in \mathcal{E}$
$\mathcal{C}$	Set of capacity markets consisting of the aFRR Up, aFRR Down, FCR-N and FCR-D	$c \in \mathcal{C}$
$\mathcal{C}^{\mathcal{D}}$	Subset of capacity markets with the down-regulating products aFRR Down and FCR-N	$c \in \mathcal{C}^{\mathcal{D}}$
$\mathcal{C}^{\mathcal{U}}$	Subset of capacity markets with the up-regulating products aFRR Up, FCR-N and FCR-D	$c \in \mathcal{C}^{\mathcal{U}}$
S^{PQ}	Index set of points (x,y) used in the linearized PQ-curve	$i \in S^{PQ}$

Parameters

Table 7: Defining the parameters used in the model.

Parameter	Description	Unit
P_t^e	Price for energy market $e \in \mathcal{E}$ in time step $t \in T$	[EUR/MWh]
P_t^c	Price for capacity market $c \in \mathcal{C}$ in time step $t \in T$	[EUR/MW]
$\overline{P_i^{\text{hyd}}}$	Discretized hydraulic power levels defining point $i \in S^{\text{PQ}}$, computed from the PQ-Curve	[MWh/h]
$\overline{P_i^{\text{net}}}$	Corresponding net electrical power delivered to the grid at point $i \in S^{\text{PQ}}$, computed from the PQ-curve	[MWh/h]
M_i	Slope between consecutive points i and i+1 on the PQ-curve, $\forall \quad i \in S^{\text{PQ}} \setminus \{\max(\mathcal{S}^{\text{PQ}})\}$	[–]
I_t	Inflow in time step $t \in T$	[GWh/h]
$C^{\max}$	Maximum production capacity	[MW]
$C^{\min}$	Maximum production capacity	[MW]
$S^{\max}$	Maximum reservoir storage capacity	[GWh]
$S^{\min}$	Minimum reservoir storage capacity	[GWh]
δ^{up}	Maximum energy demand for up regulation	[GWh]
δ^{down}	Maximum energy demand for down regulation	[GWh]
$V^{c,max}$	Maximum capacity in market $c \in \mathcal{C}$	[MW]
$V_t^{e,max}$	Maximum energy in intraday markets $e \in \mathcal{E} \setminus \{\text{DA}\}$ in timestep $t \in T$	[MWh]
S_0	Initial reservoiar level	[GWh]
A^e	Available market share in intraday markets $e \in \mathcal{E} \setminus \{\text{DA}\}$	[-]

Decision Variables

Table 8: Defining the decision variables used in the model.

Variable	Description	Domain
y_t^c	Capacity allocated to market $c \in \mathcal{C}$, at time $t \in \mathcal{T}$	[MW]
x_t^e	Production volume traded in market $e \in \mathcal{E}$, at time $t \in \mathcal{T}$	[MWh]
s_t	Storage level for the reservoir at time step $t \in \mathcal{T}$	[GWh]
p_t^{hyd}	Hydraulic power available to the turbine, i.e., power extracted from the reservoir at time $t \in \mathcal{T}$	[MWh/h]

Objective

The model aims to maximize the revenues obtained from selling energy to the DA and ID markets, in addition to capacity sold to the capacity markets. ID Buy represents the additional power in which the plant can purchase from the intraday market.

$$\max z = \underbrace{\sum_{t\in\mathcal{T}}\sum_{c\in\mathcal{C}} P_t^c\, y_t^c}_{\substack{\text{Revenue from reserve}\\ \text{capacity markets}}} + \underbrace{\sum_{t\in\mathcal{T}}\sum_{e\in\mathcal{E}\setminus\{\text{ID Buy}\}} P_t^e\, x_t^e - P_t^{\text{ID Buy}} x_t^{\text{ID Buy}}}_{\substack{\text{Net revenue from}\\ \text{energy markets}}} \quad [\text{EUR}] \tag{3}$$

Constraints

Production constraints: This constraint ensures that energy production for the given power plant respects its upper and lower production limits. Since the power plant can adjust its position through trading in the ID markets, these energy markets are also considered.

$$C^{\min} \leq \sum_{e\in\mathcal{E}\setminus\{\text{ID Buy}\}} x_t^e - x_t^{ID\,Buy} \leq C^{max}, \quad \forall t \in \mathcal{T} \quad [MWh] \tag{4}$$

Reservoir constraint: The reservoir constraint ensures that the storage in the reservoir remains between its minimum and maximum capacity, while maintaining adequate energy for obligations in the reserve markets.

$$S^{\min} + \delta^{up} \leq s_t \leq S^{\max} - \delta^{down}, \quad \forall t \in \mathcal{T} \quad [GWh] \tag{5}$$

Initial reservoir constraint: The reservoir level is initialized to a predetermined level using the initial reservoir constraint.

$$s_{t=0} = S_0 \quad [GWh] \tag{6}$$

End reservoir constraint: Load demand follows seasonal patterns and thus the end reservoir constraint ensures that the reservoir ends at the same level as it initially was.

$$s_{t=T} = S_0 \quad [GWh] \tag{7}$$

PQ-curve constraints: Ensures that the power traded in the markets does not exceed the linear interpolation of two consecutive points on the PQ-Curve.

$$\sum_{e \in \mathcal{E} \setminus \{\text{ID Buy}\}} x_t^e - x_t^{ID\,Buy} \leq \overline{P_i^{\text{net}}} + M_i \left(p_t^{\text{hyd}} - \overline{P_i^{\text{hyd}}} \right), \qquad \forall i \in \mathcal{S}^{\mathcal{PQ}},\ t \in \mathcal{T} \quad [MWh/h]. \tag{8}$$

Reservoir time step linking constraints: The reservoir storage time step linking constraint ensures that there is a physical coupling from one time step to the next, considering the inflow and discharge that have been present.

$$s_t = s_{t-1} + I_t - \frac{p_t^{\text{hyd}}}{1000}, \qquad \forall t \in \mathcal{T} \setminus \{t = 0\}, \quad [\text{GWh}]. \tag{9}$$

Down capacity bidding constraints: The maximum capacity that can be allocated in the downwards capacity regulation markets is determined by the difference between the energy sold across the various energy markets and the minimum production capacity.

$$\sum_{c \in \mathcal{C}^{\mathcal{D}}} y_t^c \leq \sum_{e \in \mathcal{E} \setminus \{\text{ID Buy}\}} x_t^e - x_t^{ID\,Buy} - C^{min} \quad \forall t \in \mathcal{T} \quad [MW] \tag{10}$$

Up capacity bidding constraints: The maximum capacity that can be allocated in the upwards capacity markets is determined by the difference between the maximum production capacity and the energy sold across the various energy markets.

$$\sum_{c \in \mathcal{C}^{\mathcal{U}}} y_t^c \leq C^{\max} - \Big(\sum_{e \in \mathcal{E} \setminus \{\text{ID Buy}\}} x_t^e - x_t^{ID\,Buy} \Big), \quad \forall t \in \mathcal{T} \quad [MW] \tag{11}$$

Maximum capacity bidding constraints: The allocation of capacity across the different capacity markets is subject to an upper bound, with values specified based on the characteristics of the modeled power plant.

$$y_t^c \leq V^{c,\max}, \quad \forall t \in T \quad \forall c \in \mathcal{C} \quad [MW] \tag{12}$$

Maximum ID volume limits: Trading volumes in the intraday market are restricted to remain within the permitted limits.

$$x_t^e \leq A^e \cdot V_t^{e,\max}, \quad \forall t \in T \quad \forall e \in \mathcal{E} \setminus \{\text{DA}\} \quad [MW] \tag{13}$$

Non-negative constraints:

$$y_t^c \geq 0 \quad \forall c \in C, \quad x_t^e \geq 0 \quad \forall e \in \mathcal{E}, \quad p_t^{hyd} \geq 0, \quad s_t \geq 0 \quad \forall t \in \mathcal{T} \tag{14}$$

7 Presenting the input data

This section presents the described data from Section 2.6 used as input for the parameters in the mathematical model. The data includes assumed plant-specific operational data and inflows, as well as market prices and tradeable volumes. Its purpose is to give the reader sufficient insight into the model inputs when interpreting the results in Section 8 and Section 9.

Parameter	**Value**
$C_{\max}$ (MW)	79.04
$C_{\min}$ (MW)	15.504
$S_{\max}$ (GWh)	101.2
$S_{\min}$ (GWh)	0
δ_{up} (GWh)	0.0175
δ_{down} (GWh)	0.0095

Table 9: Operational bounds of the hydropower plant, including generator production capacity and reservoir storage limits.

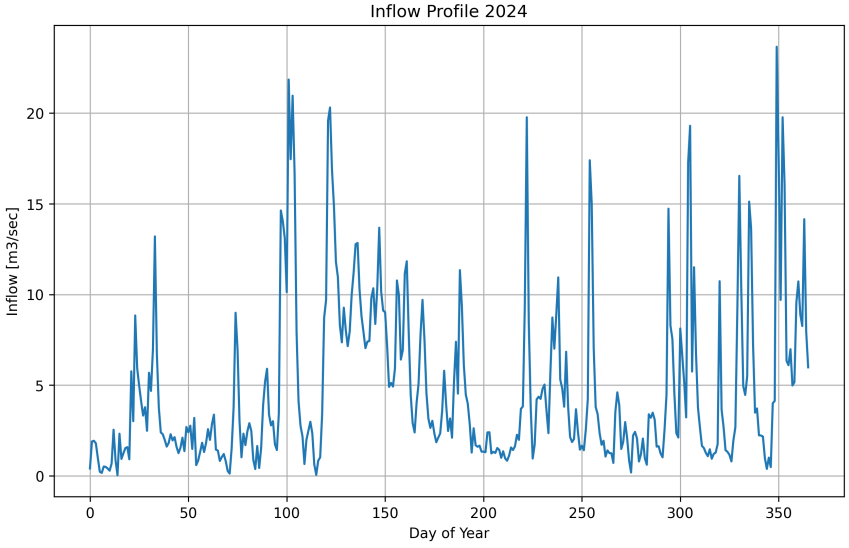


Figure 13: Inflow profile for the modelled reservoir in 2024.

Reserve Market	Mean Price (EUR/MWh)	Volume (MW)
aFRR Down	16.84	30
aFRR Up	16.05	30
FCR-N	18.80	8
FCR-D	4.36	32

Table 10: Mean capacity market prices in NO2 for 2024 and tradable volume limits.

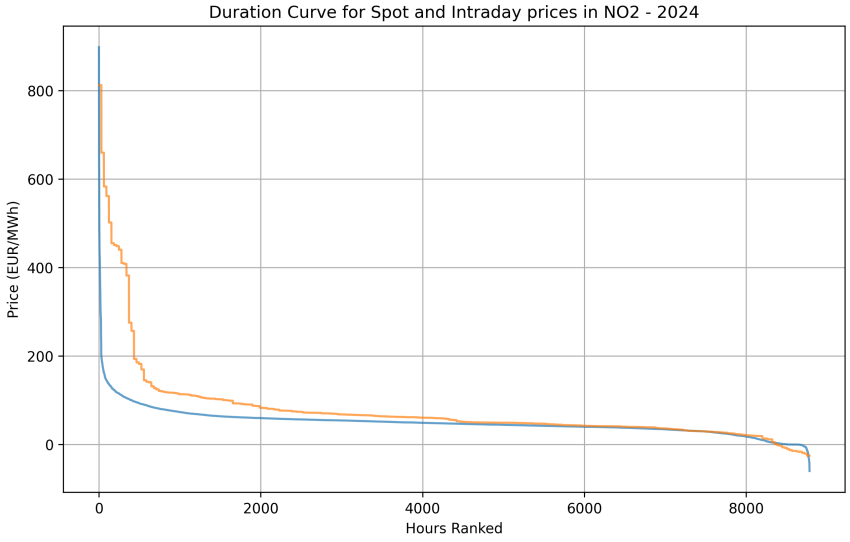


Figure 14: Duration curve for day ahead and intraday market prices in NO2 2024.

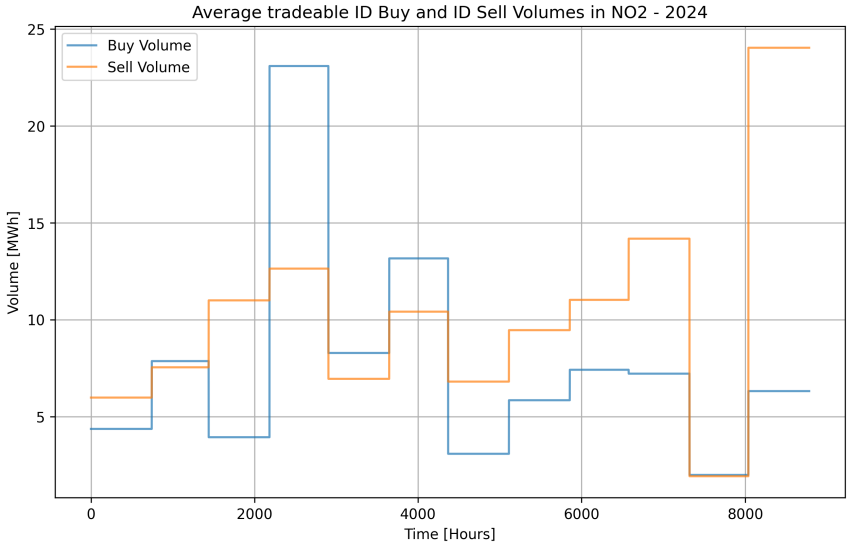


Figure 15: Maximum tradeable volumes in Intraday markets as 10% of average daily tradeable volumes.

8 Validation

The purpose of the validation is to assess whether the model's decisions align with its stated objectives and constraints. This preliminary step strengthens the credibility of the subsequent computational study.

For the validation, we restrict the planning horizon to two weeks in January. This shorter horizon allows for a clearer visual interpretation of the parameters and variables in the optimal solution. However, it also requires minor adjustments to the model. In the full-year simulation, the end-of-horizon reservoir level is required to match the initial level. This constraint is feasible for the annual model while still satisfying the minimum production requirement. In contrast, applying the same constraint to the January subset results in infeasibility. Intuitively, this is expected, as inflow levels in the beginning of January, previously illustrated in Figure 13. Therefore, for the validation exercise, the end-reservoir constraint is relaxed to $S_T = 0.75 \cdot S_0$, which is close to the highest feasible terminal storage level. Naturally, the chosen end-reservoir level influences the resulting production plan.

The validation consists of three steps. First, we assess whether the water value, represented by the dual value of the reservoir balance constraint, is reasonable. Second, we analyse the model's behaviour in the energy markets, comparing its decisions to corresponding price patterns. Finally, we examine whether the energy-market decisions appear consistent with those made in the capacity markets.

8.1 Water value

As discussed in Section 1, the water value represents the marginal income associated with storing one additional unit of water in the reservoir at the beginning of the planning horizon. In a deterministic model, this value reflects perfect foresight regarding how the additional unit can be used optimally. Running the model over the two-week validation horizon yields a constant water value of 127.271 EUR/MWh, as shown in Figure 16.

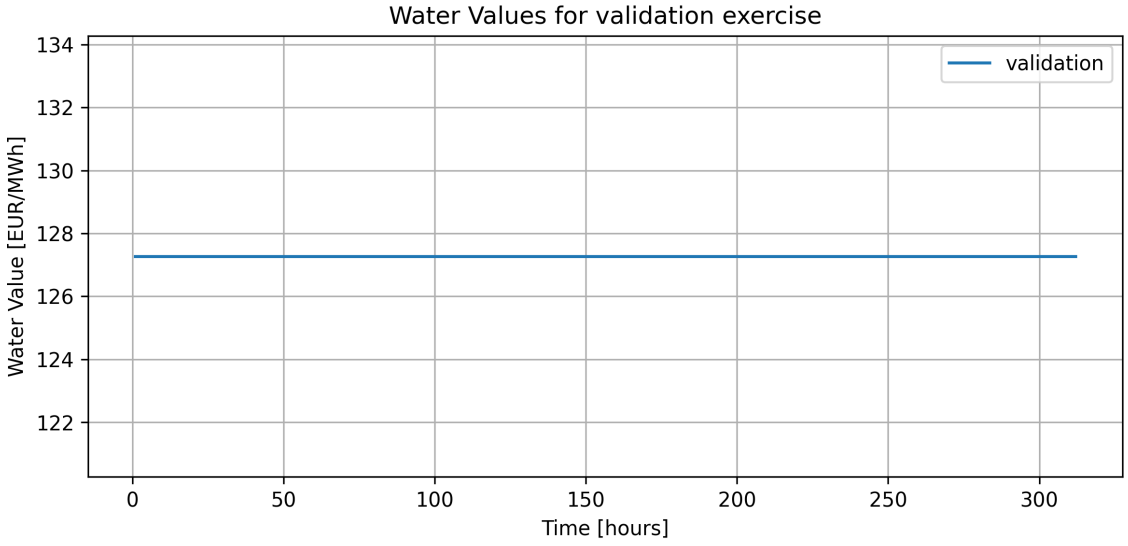


Figure 16: The water value across the planning period for the validation case.

To assess if the dual value of the reservoir balance constraint is reasonable, we apply a decomposition-based validation. First, the model is solved under the base assumptions, producing an optimal bidding strategy and a corresponding objective value, referred to as the base scenario. The model is then solved again with an additional 1 MWh of water added to the initial reservoir level S_0, which yields a perturbed optimal solution and a new objective value. Comparing these two solutions reveals how the additional unit affects both the decision variables and the objective value.

The results, shown in Table 11, confirm the expected dual value. The decomposition indicates that the additional water is optimally used in a single hour (t = 274), and that only 0.9237 MWh of it becomes economically viable due to turbine and generator efficiencies. During this hour, the marginal unit is sold in the DA-market, replacing upward capacity in the FCR-N market, and simultaneously increasing sales in the aFRR Down market. Overall, the analysis demonstrates that the model behaves as expected with respect to the water values.

Table 11: Water value revenue decomposition for the validation case.

Metric	**Value**
Objective Δ (water value) [EUR/MWh]	127.271178
Energy revenue contribution [EUR]	120.805086
Capacity revenue contribution [EUR]	6.466093
Reduced ID buy cost [EUR]	-0.000000
Decomposition total [EUR]	127.271178

t	ΔEnergy (MWh)	ΔCapacity (MW)
274	0.923728	0.923728

Market	**Marginal Contribution at t = 274 [EUR]**
DA Market	120.805086
ID Sell Market	0.000000
ID Buy Market	0.000000
aFRR Up Market	0.000000
aFRR Down Market	23.093188
FCR-N Market	-16.627095
FCR-D Market	0.000000
Total	**127.271178**

8.2 Net production in energy markets

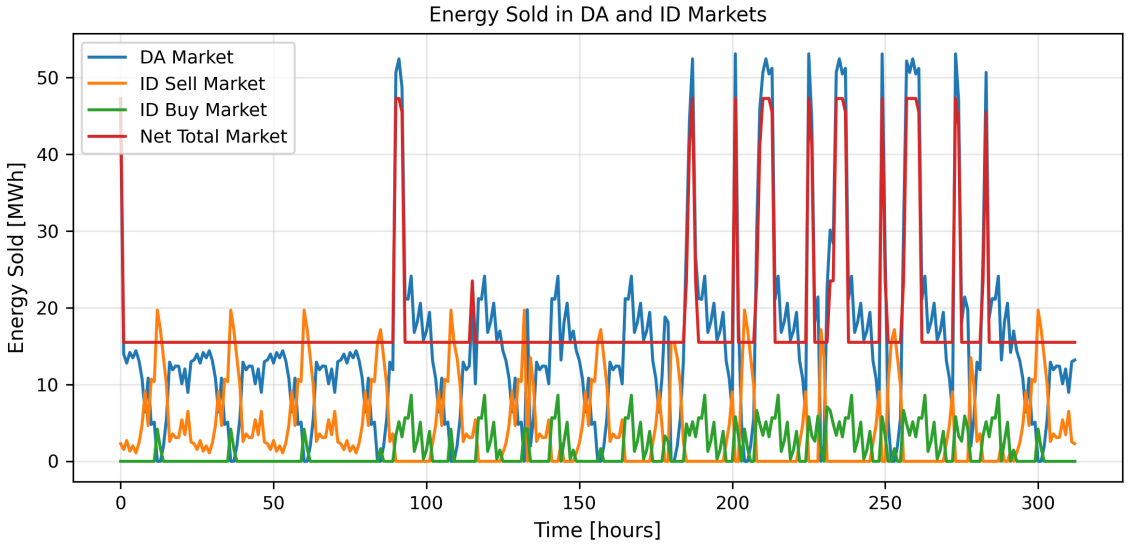


Figure 17: The energy sold across the DA and ID markets in the validation case.

In the energy markets, the model selects the most profitable combination of markets, subject to the production constraints and the reservoir balance constraint. The red curve in Figure 17 represents the net energy traded, referred to as the Net Total Market. This reflects the combined net volume traded across the DA and ID markets, and therefore corresponds to the net hourly production of the hydropower plant. For most hours, the generator operates at its minimum capacity, C_{min}, which is necessary to satisfy the terminal reservoir constraint. This behaviour is fully consistent with the intended operation of the model when accounting for both storage limitations and production requirements.

8.3 Energy market prices vs traded volumes

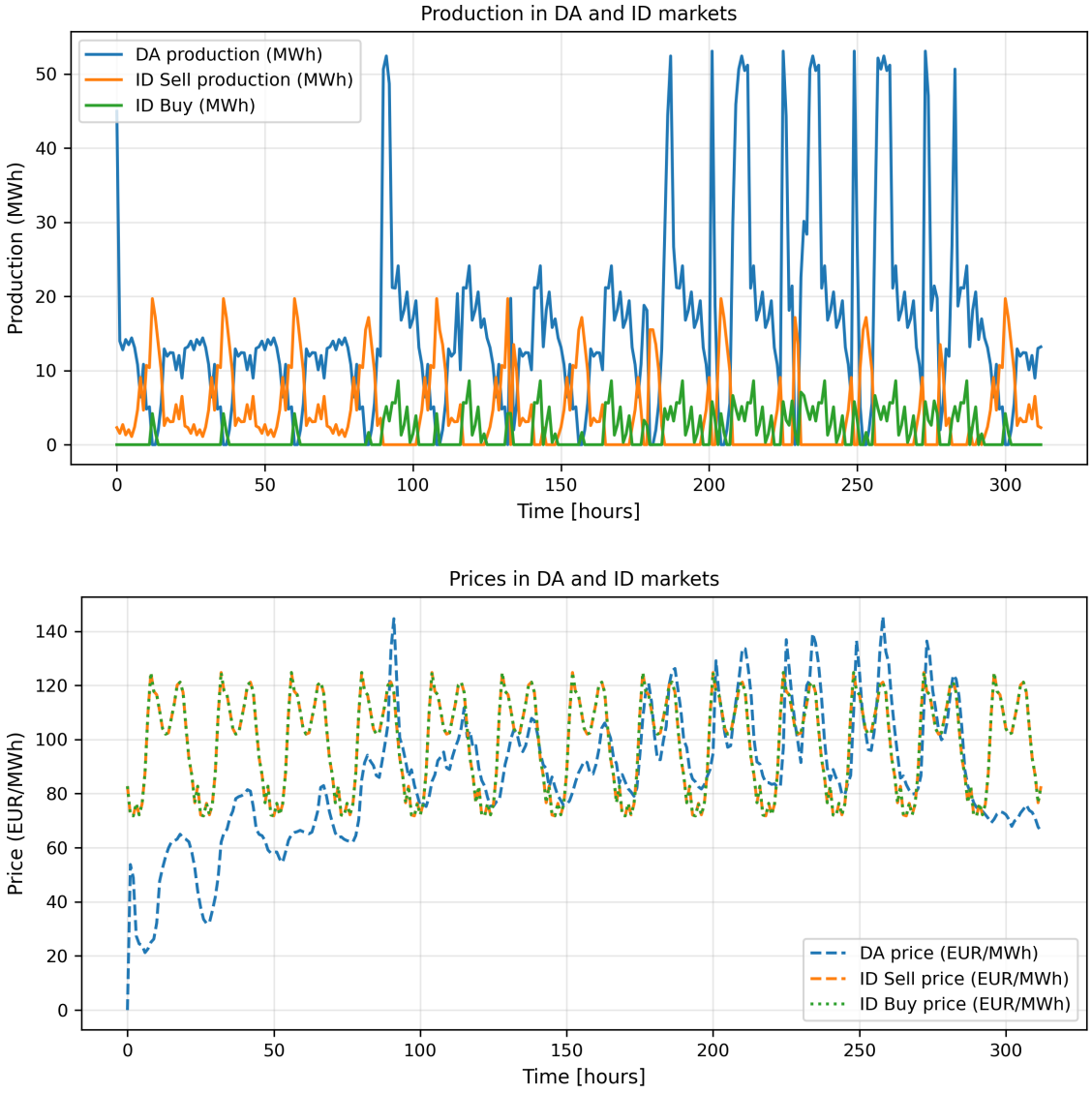


Figure 18: Production profile, and DA and ID prices in the validation case.

Extending Figure 17 to include the corresponding price series, we observe in Figure 18 how the model determines when to trade based on price fluctuations across the energy markets. Due to the terminal reservoir constraint, enforcing $S_T = 0.75 \cdot S_0$, the model

must carefully select the hours in which production exceeds the minimum operating level. We observe that this occurs predominantly in the most expensive DA hours, where the DA price also exceeds the ID prices. In other hours, where the DA price exceeds the ID price, the model exploits the arbitrage opportunity by selling more in the DA market while purchasing in the ID market such that the net production satisfies C_{min}. An additional observation is that, in most hours, the ID price tends to exceed the DA price. However, the permitted cyclic market share constrains the volume that can be traded in the ID market, limiting it to A^e of the available traded volume, which is set to 10%. Overall, the model behaves in accordance with both the market share limitations and the revenue-maximizing strategy.

8.4 Capacity market participation

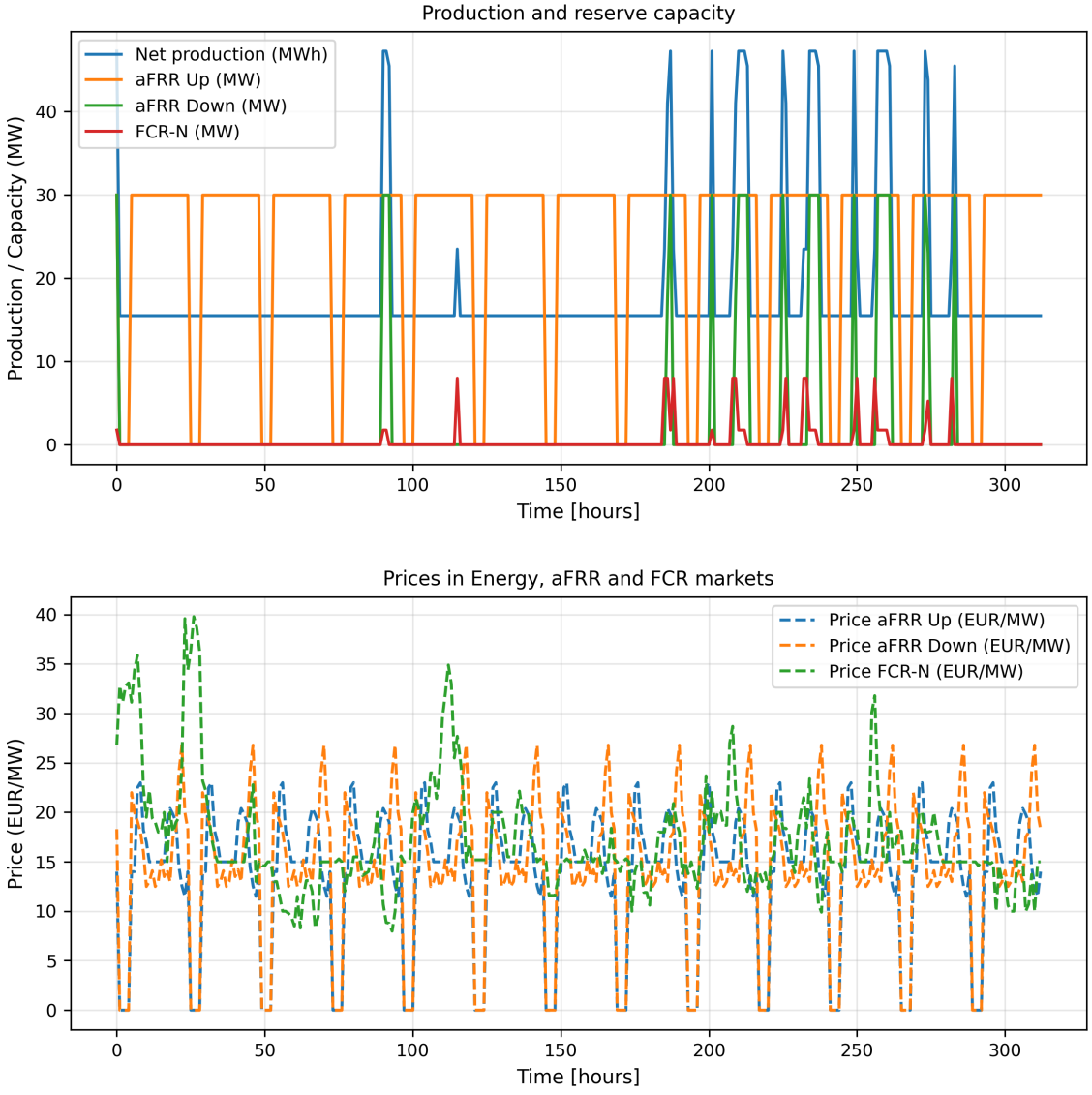


Figure 19: Production, capacity, and prices in energy and capacity markets for the validation case.

In Figure 19, we observe how the model allocates capacity according to the prices in various capacity markets, together with the net energy production. FCR-D is omitted

from the plot as all prices during this period are zero. Thus, in the validation process, the bids in this market were correspondingly zero. Figure 19 shows that as the plant seldom reaches its maximum generating capacity, the model bids the full upward regulating margin in the aFRR Up market whenever the associated price is positive. Conversely, in hours where production exceeds the technical minimum, the model allocates the excess to downward capacity products. In hours where FCR-N yields the highest marginal revenue, the model first places the maximum possible bid in this market, and subsequently allocates any remaining downward margin to the aFRR Down market. As FCR-N is a symmetric product, the available bidding volume is constrained by both the upward and downward regulating limits. We further observe that aFRR Down respects its maximum permitted bid volume. Overall, this indicates that the model behaves in accordance with the regulating constraints.

To conclude the validation exercises, it is evident that the model behaves consistently with its objective function and associated constraints. The model is therefore ready for a more thorough computational analysis of the case study. Figure 18 demonstrated that the production allocated to the energy markets is driven by the reservoir constraint and the most profitable prices. In the following computational study, the net production to the energy markets will therefore be considered as the base case.

9 Computational study

The following section presents and discusses the results from the computational study. This includes an explanation of the methodology, a brief technical presentation, and a more thorough economic interpretation of the results based on the different instances simulated in the model described in Section 6. More specifically, our computational study aims to investigate how configurations of expanded market participation perform both operationally and economically, with and without a capacity expansion. The results will be further discussed from an investor's perspective.

9.1 Explaining the model instances

Quantifying the effects of incorporating additional markets and observing how the model allocates capacity with the corresponding monetary gains requires a systematic generation and simulation of instances. These consist of a base instance and separate instances for the inclusion of aFRR, FCR, up-regulating, and down-regulating markets. All instances exclude capacity expansion for the power plant, and are depicted in Table 12

Table 12: Market inclusion for each model instance

Instance	**Day-Ahead**	**Intra-day**	**aFRR up**	**aFRR down**	**FCR-N**	**FCR-D**
BASE	✓	✓	–	–	–	–
BASE + aFRR	✓	✓	✓	✓	–	–
BASE + FCR	✓	✓	–	–	✓	✓
BASE + Up Markets	✓	✓	✓	–	✓	✓
BASE + Down Markets	✓	✓	–	✓	✓	–
Total	✓	✓	✓	✓	✓	✓

9.1.1 Connecting water values and the reservoir level

Before proceeding with the computational study, the base instance is used to illustrate how the water value behaves in relation to the evolution of the reservoir level.

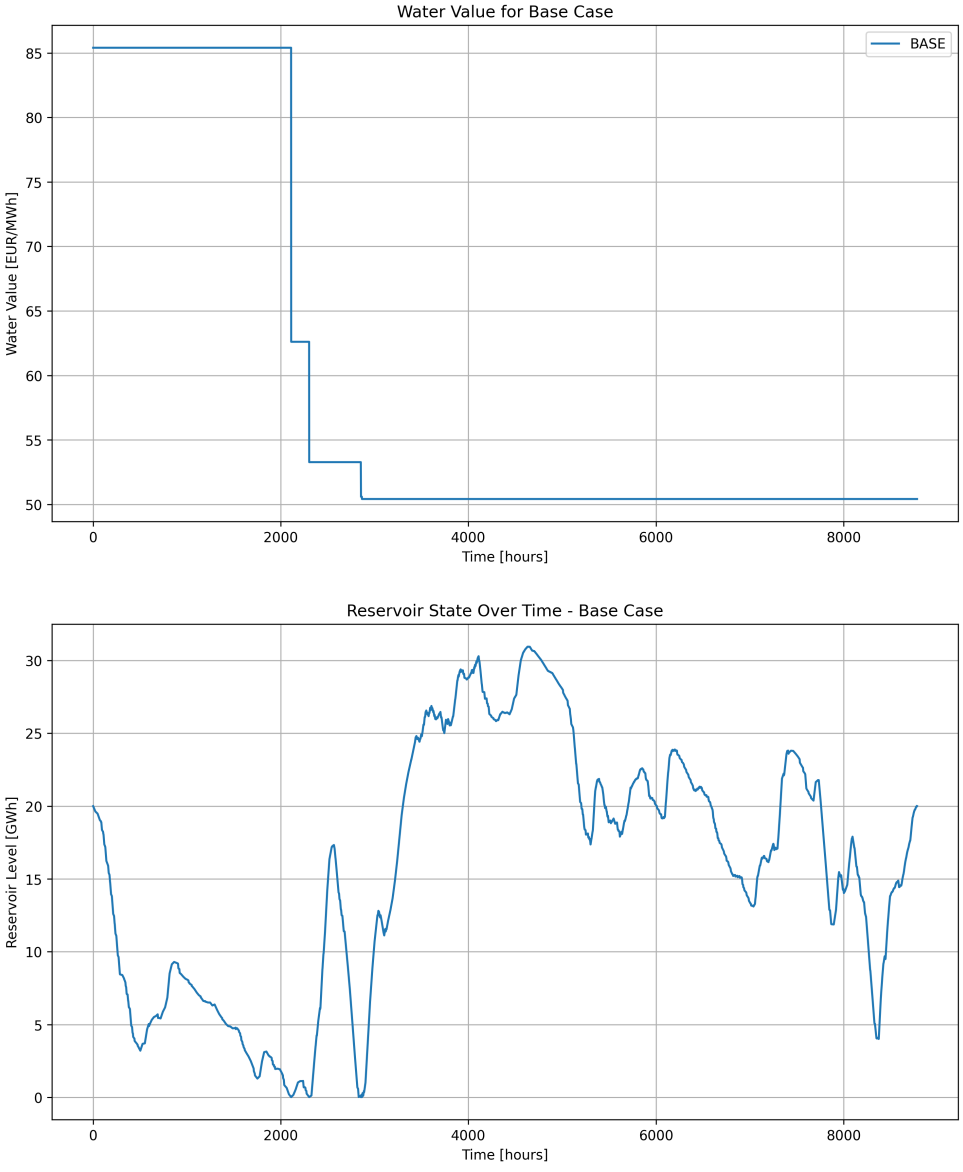


Figure 20: Water value and reservoir level for the full-year base instance.

Examining the water value in Figure 20 and comparing it with Figure 16, it can be observed that the former exhibits several discontinuities, rather than a single constant value as in the latter. When considering these discontinuities in the context of Figure 20, it becomes apparent that they occur whenever the reservoir is fully depleted. Because the model has perfect foresight regarding prices and inflows, the production schedule is optimized to maximize certain revenues. Consequently, the reservoir is emptied whenever this constitutes the optimal strategy. Between such depletion events, there is a marginal hour in which allocating one additional unit of water to the market would be optimal. As the model is fully deterministic, it is therefore reasonable that the water value evolves as a strictly decreasing function. Each depletion point represents a change in the opportunity cost of stored water, which induces a new marginal valuation segment. In the remainder of the computational study, this insight can be recalled whenever water values are presented.

9.2 Technical execution

The model was solved using Python version 3.12.10 and is implemented in the open-source optimization modelling language Pyomo. As solver to the optimization model, Gurobi version 12.0.3 optimization software was used. The model was run on a computer with an AMD Ryzen 5 5600H with Radeon Graphics at CPU at 3.301 GHz and 16 GB of installed memory (RAM). As the model is kept linear and the number of variables and constraints are rather low, further technical analysis of run time and number of iterations will not provide results relevant to the further discussion.

9.3 Analyzing operational and economic performance

The defined instances enable the subsequent computational study of the hydropower plant's operational and economic performance from an investment perspective.

9.3.1 Operational performance

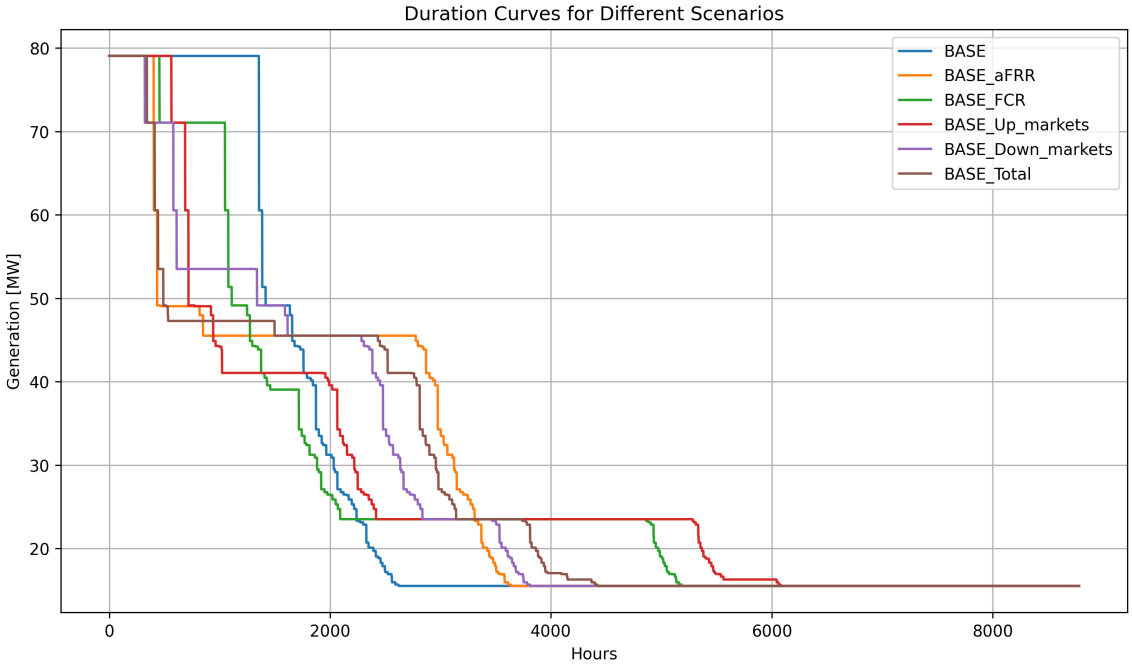


Figure 21: Duration curves across the various instances.

A duration curve, as presented for the different instances in Figure 21, illustrates how the generation capacity of the plant is utilized throughout the year, sorted from the highest to the lowest output levels. It appears that excluding capacity market participation leads to a greater number of hours in which the power plant operates at maximum production capacity, relative to configurations that include reserve provision. Moreover, the production level appears closely linked to the amount of capacity allocated to the respective reserve markets. For example, the aFRR instance has access to larger capacity market shares than the FCR instance. This is reflected in periods of operation where the plant can simultaneously offer its maximum upward and downward regulating capacity.

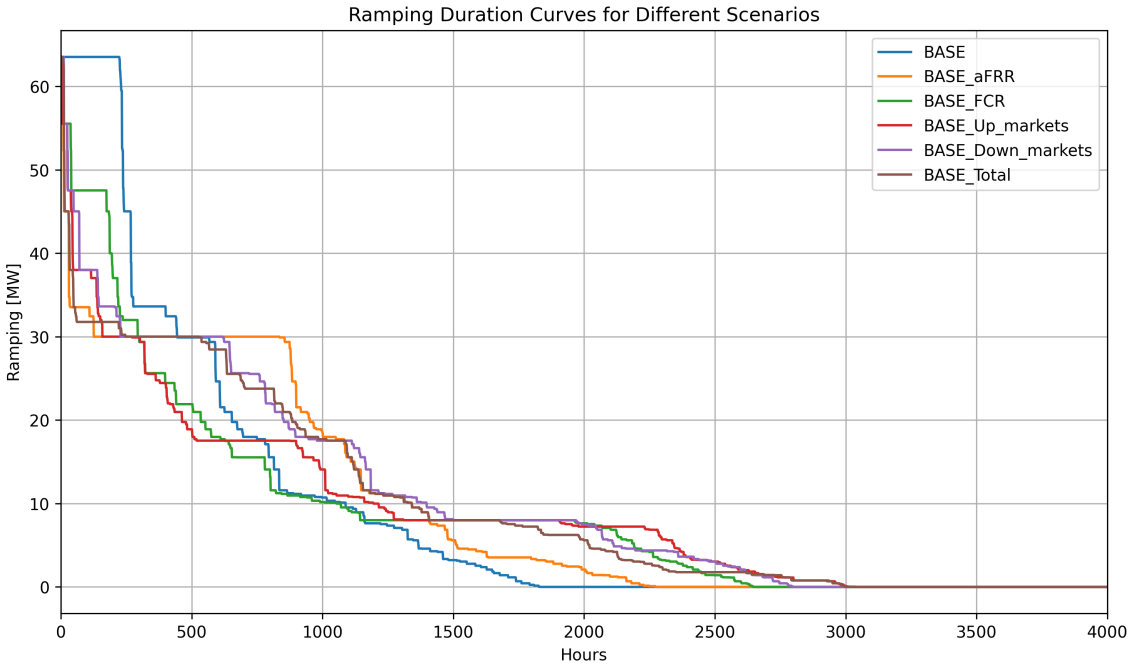


Figure 22: Ramping duration curves across the various instances.

The characteristics of the duration curve can also be used to examine how different market configurations affect the ramping behavior of the power plant. In this context, ramping refers to all hours in which the production level in hour t changes the next hour t+1. From Figure 22, it can be observed that the base instance exhibits a higher frequency of maximum ramping events. As the model seeks to maximize revenue, the production level responds accordingly by exploiting price fluctuations in the energy markets. When capacity markets are introduced, the overall magnitude of ramping generally decreases, while the number of hours in which ramping occurs increases. Since lower average production levels enable greater bidirectional flexibility, it is reasonable that the ramping magnitude decreases across configurations that include reserve market participation. This change in operational pattern is particularly pronounced in the configuration involving down-regulating markets. In this case, the model is incentivized to maintain a sufficient and stable production profile to offer down-regulating capacity.

Conversely, exploiting short-term price differences across markets, and thereby inducing frequent ramps, is most apparent in the up-regulation configuration, as well as in the total configuration. Significant ramping from one production level to another

can lead to deviations from the optimal efficiency operating point and pose additional mechanical stress on the unit (Liu et al., 2016). In this respect, participation in capacity markets may be beneficial. It should be noted, however, that accounting for activation would likely lead to a different ramping profile.

9.3.2 Economic performance

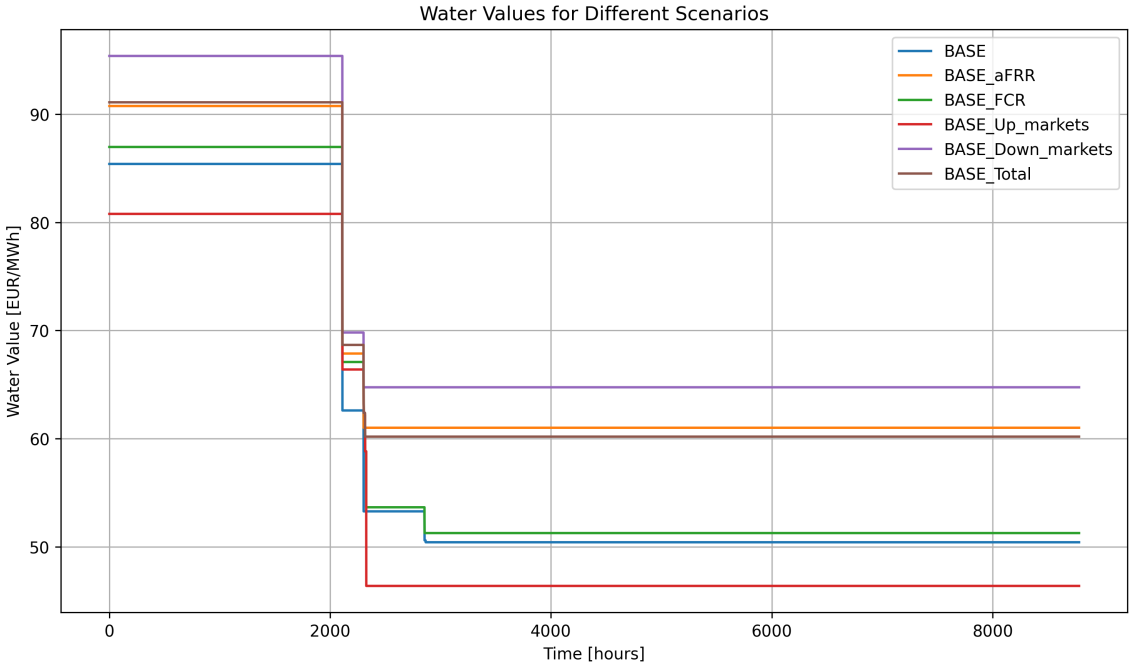


Figure 23: The water values across the various instances.

Presenting the water values for the different model instances in Figure 23 indicates that the marginal value of an additional unit of stored water is highest in the down-regulating instance, and lowest in the up-regulating instance. Interpreted within the model context, this suggests that participation in down-regulating reserve markets is more profitable than participation in up-regulating markets. One additional unit of water can be sold in the Day-Ahead market while simultaneously being offered as down-regulating reserve capacity, effectively yielding a double payoff to the producer. In contrast, in the up-regulating setup, the same marginal unit must be allocated to the DA market, displacing the marginal unit in the up-regulating reserve market.

Thus, it cannot generate revenues in both markets simultaneously. The lower water value implies that storing water for future use is considered less valuable, as the incentive to retain water diminishes in the absence of downregulating markets.

Extending the discussion from Section 9.3.1, the plotted water values also highlight a relationship between market share limitations in the capacity markets and the number of reservoir emptying cycles. The instance with only access to the FCR markets has a relatively low market share and many periods with zero capacity prices. Consequently, the producer faces fewer incentives to defer production into later hours and therefore tends to empty the reservoir more often, a behaviour similar to the base case. From a flexibility perspective under uncertainty, such reservoir depletion may be undesirable, implying that participation in downward reserve markets encourages operating patterns that preserve flexibility while increasing revenues.

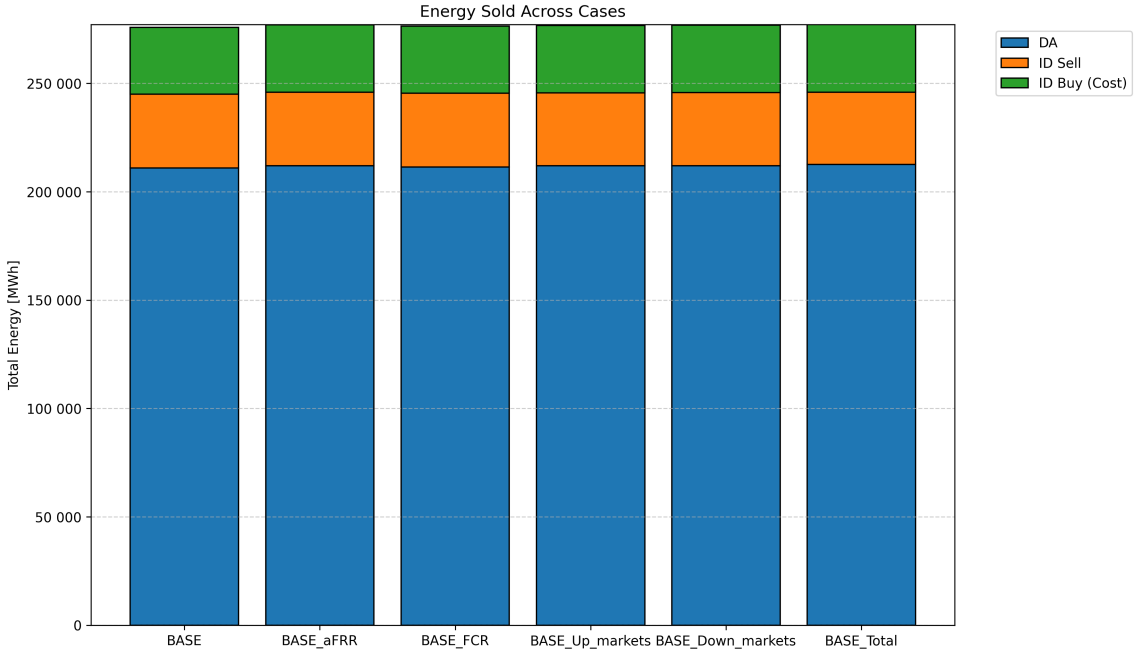


Figure 24: Energy traded in the markets for the various instances.

The foregoing results have shown that distinct operational patterns emerge depending on the market configuration under consideration. While these differences influence the hourly production profile and thus the reservoir trajectory, Figure 24 demonstrates that the annually accumulated net energy production is nearly identical across

all instances. This indicates that the inclusion of capacity market participation does not materially affect the total energy generated over the year. Given that the energy markets remain the most profitable outlet for the plant, it is reasonable that the model seeks to maintain a stable aggregated level of annual energy production.

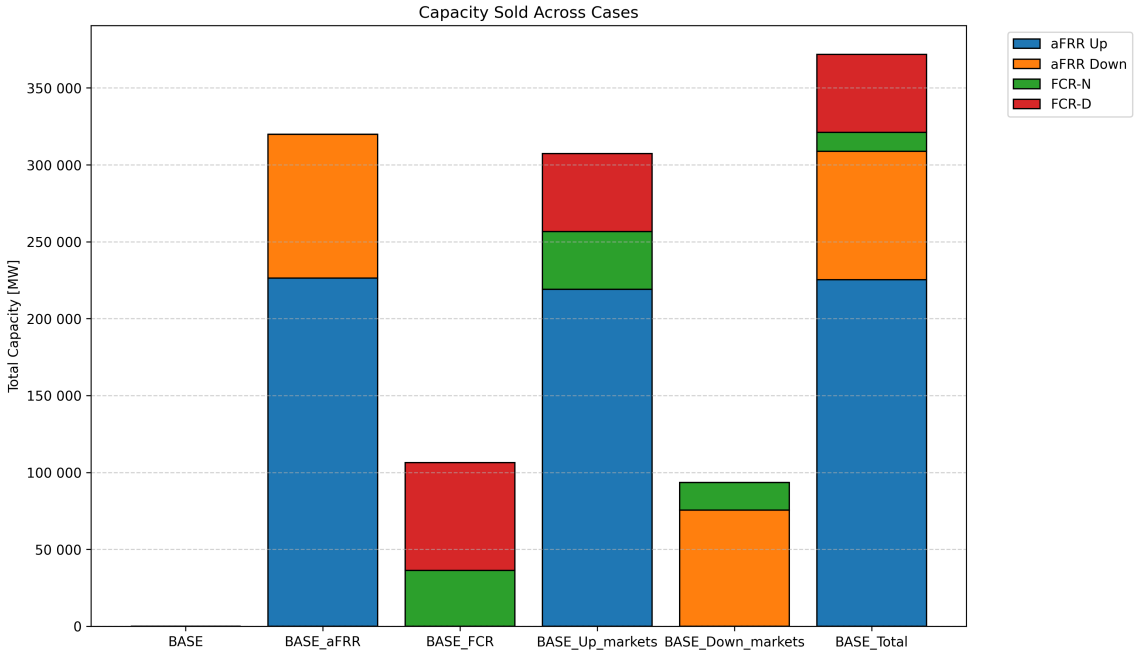


Figure 25: Capacity sold to the markets across the various instances.

Examining the corresponding aggregated contributions to the capacity markets in Figure 25 enables a discussion on which products are preferred, and why. The most noticeable observation is the dominance of aFRR Up whenever this market is included. In the context of a revenue-maximizing strategy, this dominance can be attributed to comparatively lucrative prices observed in this market, which frequently coincide with favourable prices in the energy markets. These synergies appear to be stronger for aFRR Up than for aFRR Down, despite both markets having access to the same amount of available capacity. Additionally, since the power plant is restricted to volume limitations of the reservoir, it would have a volume bias towards bidding up-regulating capacity whenever the plant does not produce at maximum production capacity.

Furthermore, the contribution to FCR-D is larger than the contribution to FCR-N.

Given that the former has four times the available market share of the latter, this outcome seems reasonable, even though the FCR-D price is equal to zero for most hours. Consolidating these results for the total configuration suggests that the model tends to prioritise aFRR Up, followed by aFRR Down, then FCR-D, and finally FCR-N. This hierarchical bidding pattern reflects the combined effect of price signals and available market capacity, of which the latter is logically the most decisive factor.

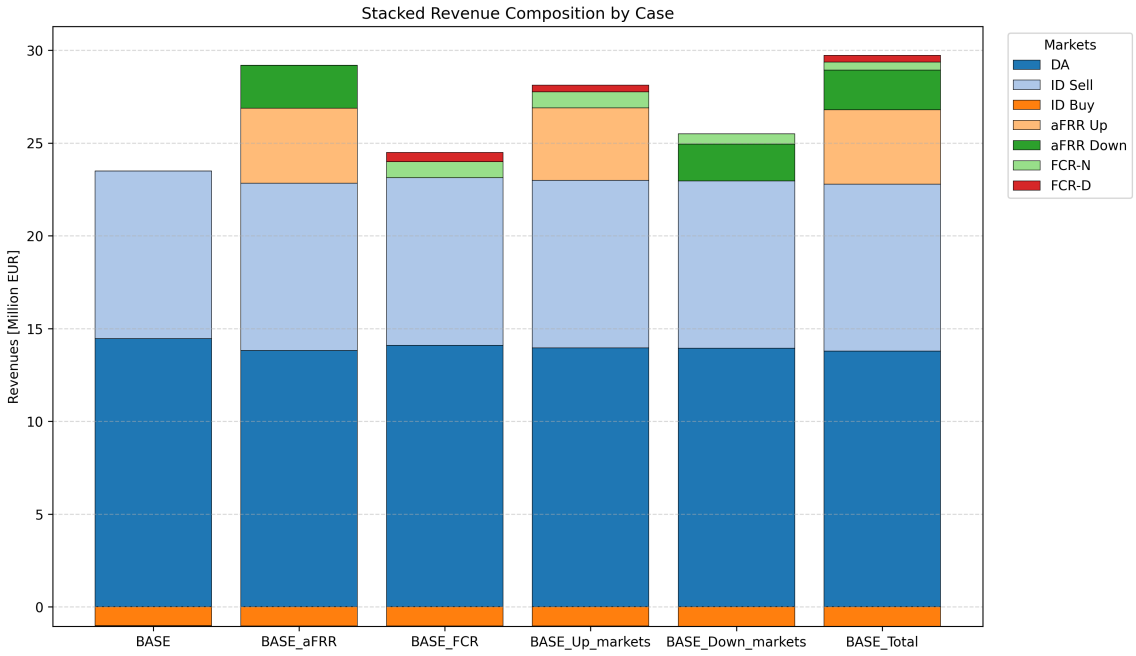


Figure 26: The revenues obtained from the various instances.

Figure 26 illustrates the preceding discussion on the economic performance analysis. While the introduction of capacity market participation leads to a slight reduction in energy-market revenues, the latter remain relatively stable across the different configurations. Hence, participation in multiple capacity markets provides additional value on top of the energy that is produced. The extent to which capacity is allocated to each reserve product is determined by the available market capacity, the associated price levels, and the production level, which together form the basis for the prioritization discussed above.

To highlight these contributing factors, the annual revenues can be expressed in terms of total revenues per MW of annual tradeable capacity. These values can be

compared to the mean prices used in Section 7. For practical purposes, we restrict this comparison presented in Table 14 to the total instance and the annual mean prices in NO2.

Market	**Total revenue [EUR]**	**Revenue per MW [EUR/MW]**	**Mean price [EUR/MW]**
aFRR Up	4 015 494.74	15.24	16.05
aFRR Down	2 144 774.91	8.14	16.84
FCR-N	427 313.19	6.08	18.80
FCR-D	370 761.50	2.27	4.36

Table 14: Annual revenue per MW and mean prices for the individual reserve markets in NO2.

Examining Table 14 illustrates the volume bias through the aFRR Up annual price per MW. Further, it shows the aFRR markets are favoured due to their large tradeable market shares and price patterns compared to the FCR markets. It also important to remark that the resulting values are also influenced by decisions made in the energy markets as this influences the available capacity to bid into the capacity markets. Consequently, the revenue per MW differs from the mean price, reported in Table 14.

9.4 Modelling revenues with capacity expansions

The computational study thus far has demonstrated that participation in multiple markets alters the operational pattern and generates additional value for the hydropower producer. A central strategic question is whether this additional value can justify investments in expanding production capacity. Consequently, the remainder of the computational study examines how different levels of capacity expansion influence the expected cash flows associated with multi-market participation. In the model, this is implemented by adjusting the maximum generation capacity and the maximum capacity that can be offered in the respective capacity markets.

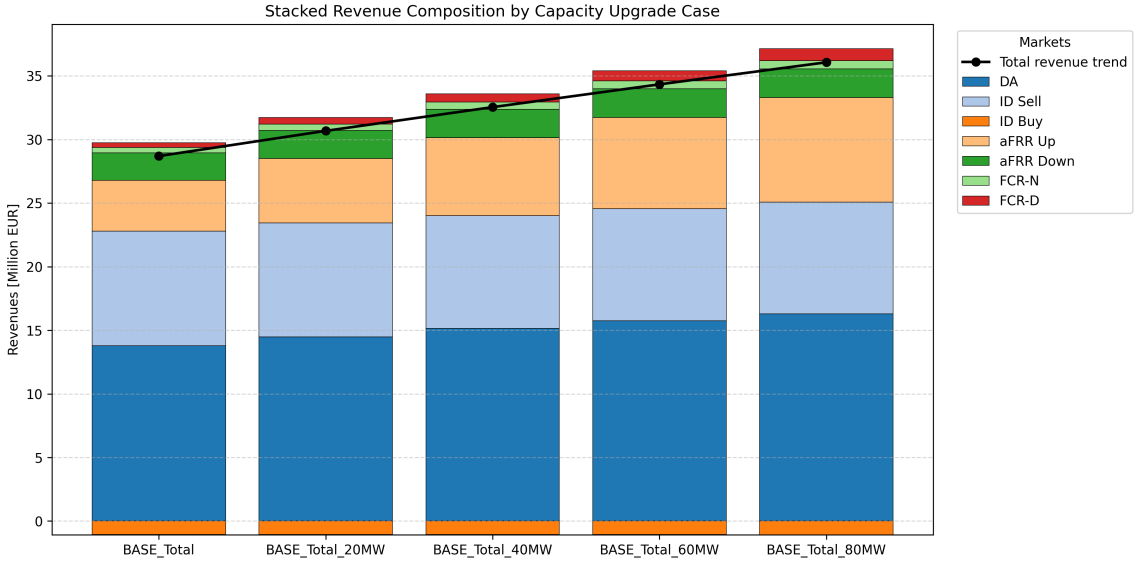


Figure 27: Revenues obtained from base case and the various capacity expansion cases.

From Figure 27, it can be observed that the revenue effect of an additional x MW of capacity expansion exhibits a weakly concave pattern. This suggests that the marginal benefit of additional capacity upgrades declines as the installed capacity increases. With a given capacity upgrade, the maximum level of market participation in the reserve markets scales proportionally with the increase in installed capacity. However, as the volume limits in the intraday market are fixed, this market does not provide an income source that can be proportionally leveraged. The diminishing revenues are in accordance with diminishing marginal utility presented in economic theory (Lin & Peng, 2019). Moreover, the figure indicates that the underlying revenue composition remains largely unchanged across the scenarios. Since the power plant acts as a price taker, the additional capacity primarily enhances its ability to exploit high-price hours across the various markets.

9.4.1 Investment analysis

When evaluating the long-term effects of a potential capacity expansion, it is relevant to conduct an NPV analysis, as outlined in Section 3.2.4. As these investments are inherently complex, assumptions regarding expected expenditures are required

and subsequently compared with the incremental revenues enabled by the expanded capacity. For this analysis, it has been assumed that the capacity expansion has an economic lifetime of 40 years and that annual O&M costs correspond to 2% of the initial investment cost (IRENA, 2012). It is further assumed that the investment cost is incurred one year prior to the commencement of operation. The annual revenues associated with the incremental capacity expansion is presented in Figure 28.

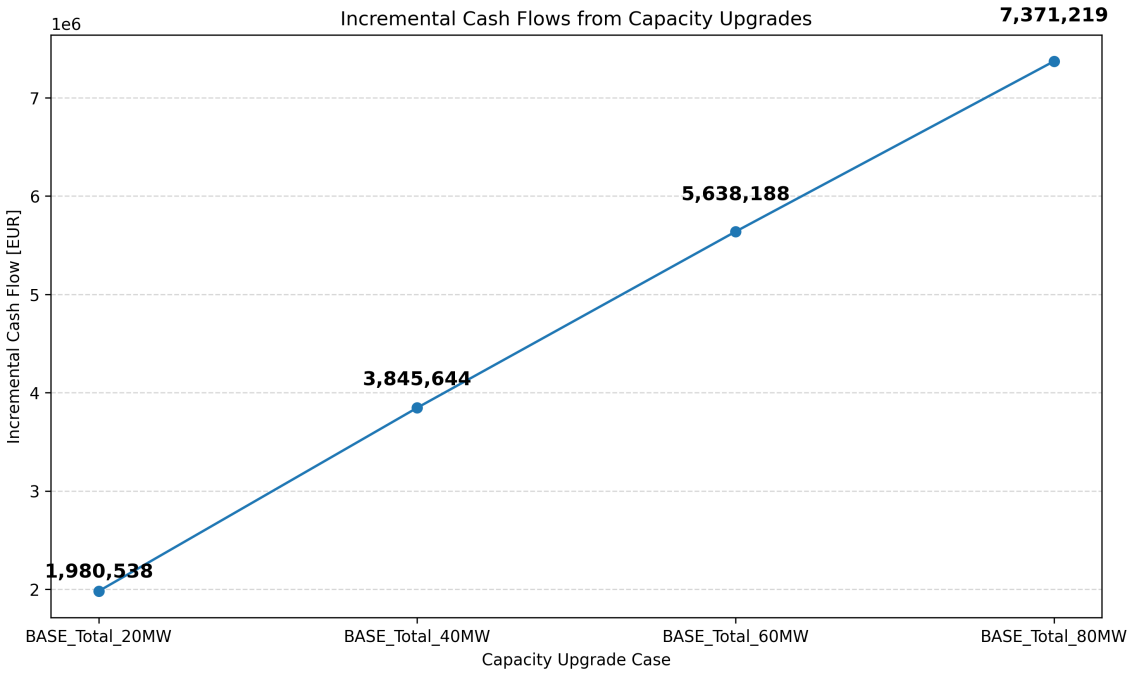


Figure 28: Incremental annual revenues obtained from the capacity expansions.

Using the NPV expression in Equation 2, the incremental cash flows can be assessed relative to the investment costs and the annual O&M expenses. Given the considerable uncertainty associated with investment costs, it is useful to determine the maximum investment cost per MW that still yields a net present value equal to zero. This assessment is carried out across a range of discount rates and for the full set of cash-flow outcomes associated with the different capacity-expansion alternatives. By solving the NPV equation for these combinations of cash flows and discount rates, an estimate of the maximum economically justifiable investment cost per MW is obtained.

Table 15: Per-MW investment cost required to achieve NPV = 0 for various discount rates r, and various capacity upgrade cases

Capacity upgrade [MW]	$r = 0.03$ **[EUR/MW]**	$r = 0.07$ **[EUR/MW]**	$r = 0.10$ **[EUR/MW]**
20	1 565 336	1 042 288	809 973
40	1 519 720	1 011 914	786 370
60	1 485 398	989 061	768 610
80	1 456 478	969 804	753 645

When these investment cost levels are compared with the per-MW benchmarks referenced in Section 3.2.4, ranging from 625,000 to 1,250,000 EUR/MW, the results suggest that the proposed capacity upgrades could be economically feasible. At the lower end of the investment range, the upgrades are expected to be profitable across all three discount rates and for all incremental cash-flow scenarios considered. At the upper end of the range, however, profitability becomes more sensitive to the required rate of return, and the highest discount rate may only be justifiable at around 3%. While these results indicate that investing in increased power capacity might be profitable, it is important to emphasize the uncertainty regarding these cash flows, as the investment analysis is based on a single year of historical data.

10 Concluding remarks

This project has examined electricity markets, recent price developments, and key modelling considerations relevant for hydropower systems. A central finding from the literature review is that existing studies largely lack analyses of multi-market participation within a mid-term planning context. To address this gap, the project has focused on a deterministic mid-term hydropower planning problem that explicitly allows participation across multiple energy and capacity markets. The purpose has been to analyse how the model allocates capacity across markets, how this allocation influences the operational behaviour of a hydropower plant, the economic value generated by such multi-market participation, and how these effects should be interpreted in relation to investment decisions in increased capacity.

Operationally, the results indicate that participation in capacity markets leads to a lower utilisation of turbine capacity, as large parts of the available capacity is reserved for reserve market participation. At the same time, the model exhibits a higher number of ramping hours, although with smaller amplitude variations. These findings imply that the hydropower plant engaged in multi-market participation tend to operate less aggressively in a market situation without considering activation. This can, for several hours, reflect a strategic positioning for both upward and downward regulation. Moreover, as the plant undergoes fewer transitions between near-minimum and near-maximum production levels, the mechanical wear is expected to be reduced.

Economically, the results show that participation in capacity markets can lead to fewer reservoir drawdowns and that the model places a relatively high value on down-regulation, as this may generate double revenue streams. At the same time, there is a volume bias towards up-regulation. While energy markets remain the dominant source of revenue, the tradable volumes and price patterns in aFRR favour participation in this market rather than FCR-N, despite FCR-N exhibiting the highest average prices. These findings highlight the key drivers influencing the revenue potential of capacity market participation. Further, it illustrates how the reservoir constraints may favour up-regulation, while the input data incentives down-regulation as energy and down-regulating prices exceeding up-regulation prices alone. This reveals important trade-offs inherent in multi-market optimisation.

From the investment perspective, the analysis indicates diminishing marginal benefits from increasing installed capacity. Nevertheless, the additional value generated through capacity market participation can justify costly investments in increased capacity, also under higher discount rates. The implications of these results sug-

gest that capacity upgrade investments can be economically viable, but uncertain future prices and investment costs necessitate conservative assumptions to support well-informed and robust investment decisions.

Overall, this work provides insights into both the technical and economic characteristics of electricity markets, as well as key modelling considerations relevant to hydropower applications. The findings establish a strong foundation for further model development, with the potential to support more robust decision-making for hydropower producers. Future work can include more detailed representations of stepwise dynamics of the power markets, explicit modelling of reserve activation, incorporation of uncertainty, refined cost modelling, and the application of advanced solution methodologies as the problem formulation increases in complexity.

11 Future Research

There are several promising directions for future research, such as model extensions and the application of more advanced solution techniques as the problem becomes increasingly complex.

Possible model expansions

Stochasticity
A natural first step would be to incorporate stochasticity in energy and capacity market prices. Assuming perfect foresight of future market conditions is optimistic and can lead the model to operate the hydropower plant closer to its physical and operational limits. Additionally, having perfect knowledge of future inflows makes the model unrealistically confident in its water management decisions. Lastly, traded volumes in the ID market can also be regarded as stochastic. Given that the model limits trading to the realised hourly traded volumes, this constraint may also be treated as a stochastic parameter.

Building on the lack of modelling uncertainty, the model currently exploits arbitrage opportunities arising from differences between DA and ID prices. With greater uncertainty in ID and DA prices, the model would likely refrain from selling large quantities of energy in the DA market and subsequently buying them back in the ID market, as this strategy would become riskier under uncertain price conditions. Changes in the water values are also expected to change relative to the baseline water value trajectory, which exhibits a stepwise pattern over the planning horizon. Finally, reservoir levels would likely appear more stable, as the model would be less inclined to operate the power plant aggressively early in the planning horizon to maximize revenues at specific high-price hours.

When incorporating stochasticity, modelling the decision hierarchy becomes a natural point of focus. The timing of information availability and bid submission deadlines provides the basis for constructing multi-stage formulations. Under such a framework, the problem could be further developed, allowing the stepwise requirements of the power market to be represented more realistically. To the best of the author's knowledge, such an extension would provide new and valuable insights, particularly in a multi-market context.

Cost and degradation
Another interesting aspect concerns the influence of generator start-up and shut-down costs, detailed O&M expenses, taxes, and degradation effects. Since the current model only accounts for operational revenues, incorporating these components into the optimization model would possibly lead to a more realistic representation of actual operational decisions. However, the inclusion of start-up/shut-down costs is likely to increase computational complexity. Introducing start-up and shut-down costs could necessitate the use of binary variables to indicate whether the generator is started or shut down in each hour, thereby adding complexity to the optimization problem, which must be represented over the entire planning period of one year.

In contrast, taxes, O&M costs and degradation effects could be incorporated without necessarily increasing computational complexity, although they would require more detailed data to produce a realistic estimate of profitability. Additionally, it is worth emphasizing that these components, particularly taxes, are important considerations for power producers and may enhance the study's relevance to the industry.

Reserve activation
Regarding more detailed modelling of the reserve markets, there also exists a clear potential to incorporate reserve activation in a manner that is both suitable and computationally efficient. The approach proposed by Chazarra et al. (2016) can be regarded as a simpler, yet suitable modelling alternative.

Transition to 15-minute time resolution
Lastly, another potential direction for future research is to move from an hourly to a 15-minute time resolution. However, the limited availability of high-resolution data may pose a challenge. Moreover, such a refinement would increase computational complexity, as the planning horizon would effectively quadruple and each decision variable would need to be represented at the quarter-hour level throughout the entire planning period.

Possible solution methods

As the model becomes more complex, it is relevant to adapt solution techniques to ensure that the problem can still be solved within a reasonable time. Based on the literature review, there appears to be potential for incorporating a stochastic dynamic programming framework into the presented optimization problem. This may be achieved by decomposing the problem formulation into subproblems, where the

stochastic parameters are treated deterministically within each respective timeframe.

Helseth et al. (2017) provide valuable insights into this modelling framework and can serve as inspiration for future work. Incorporating this type of framework alongside a more detailed representation of the various reserve markets would, to the best of the authors' knowledge, offer new and valuable insights for hydropower producers. Consequently, the study is relevant to the industry at a time when capacity upgrades for participation across markets are an increasingly important topic.

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Appendix

A Analysing capacity market prices

The aFRR market

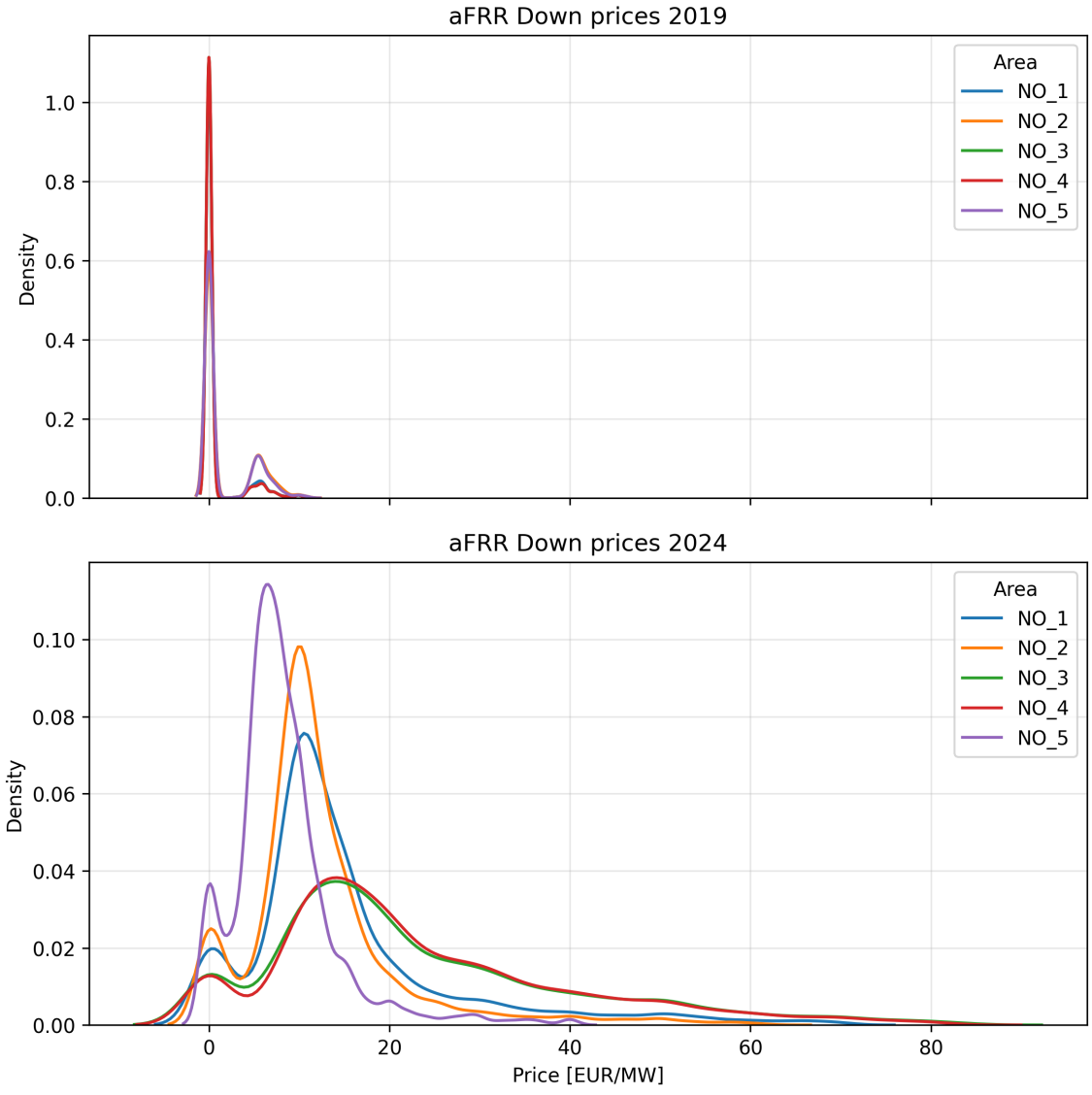


Figure 29: aFRR Down price distribution (2019 vs 2024)

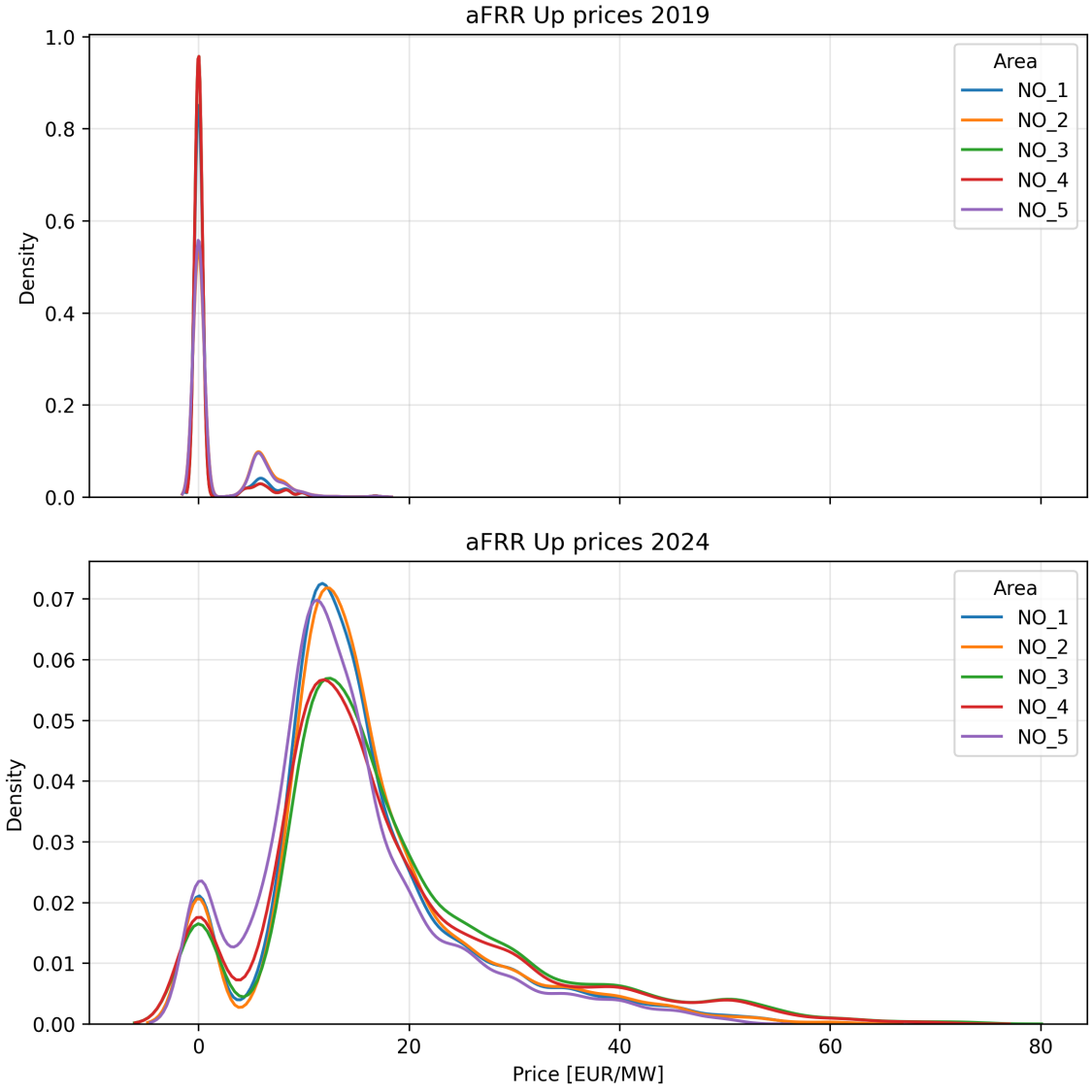


Figure 30: aFRR Up price distribution (2019 vs 2024)

The mFRR market

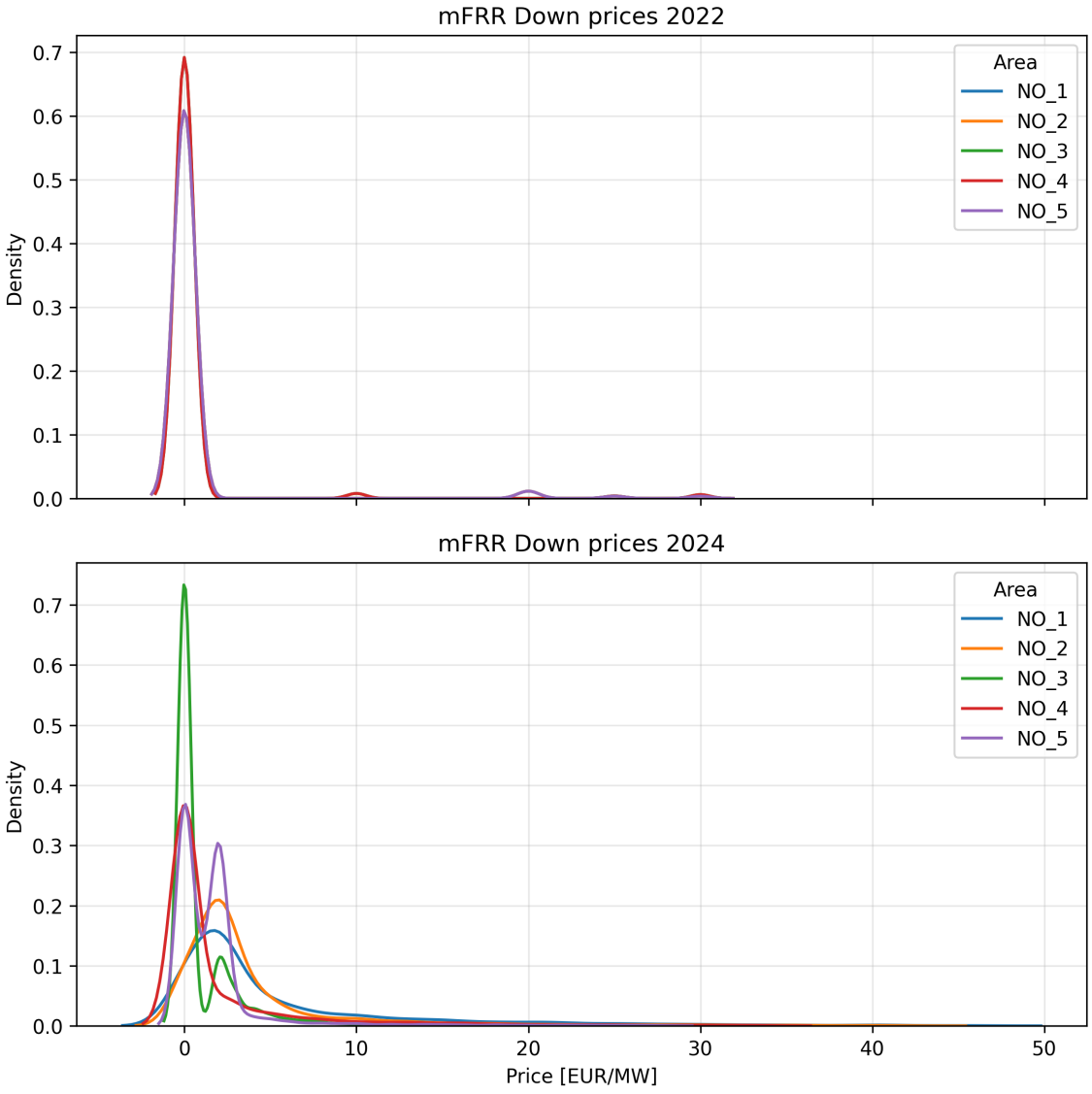


Figure 31: mFRR Down price distribution (2019 vs 2024)

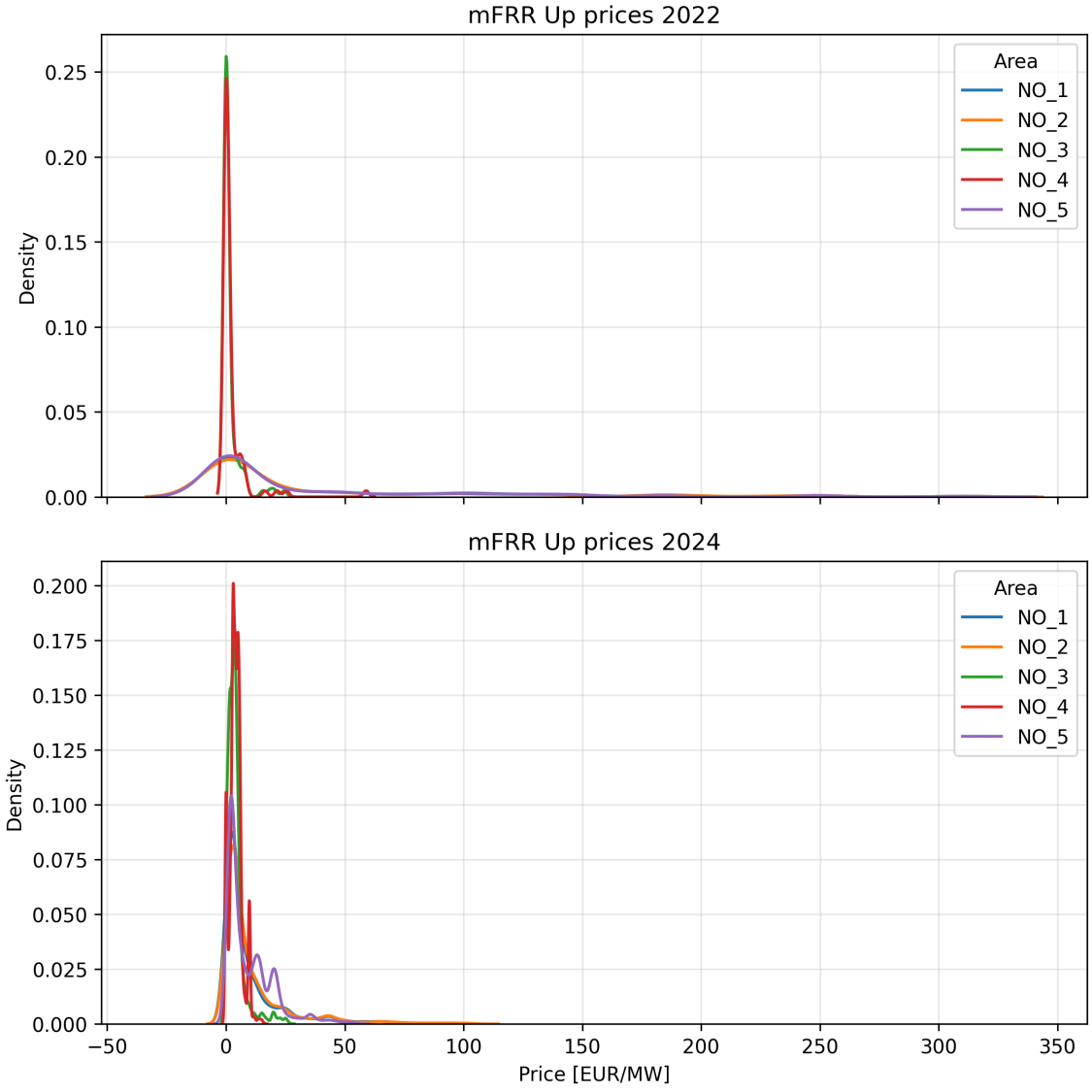


Figure 32: mFRR Up price distribution (2019 vs 2024)

The FCR-D market

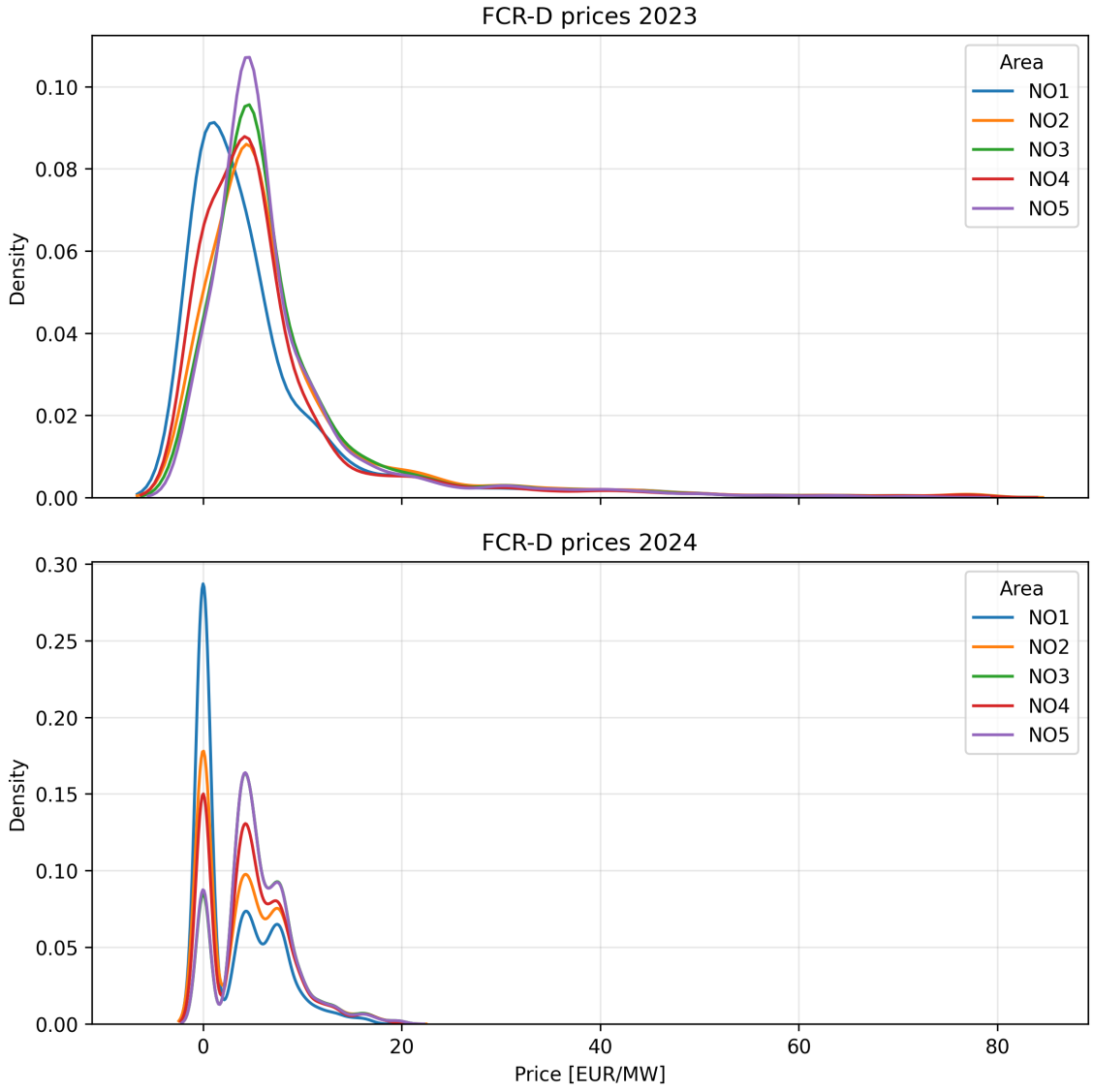


Figure 33: FCR-D price distribution (2023 vs 2024)

B Turbine Efficiency Curve

The turbine efficiency is defined as a function of the water discharge Q [m^3/s]. Table 16 lists the measured efficiency values used to construct a piecewise–linear approximation for the optimization model.

Table 16: Turbine efficiency as a function of discharge

Index	**Discharge** Q [$\mathbf{m^3/s}$]	**Efficiency** η_{tur} [%]
1	0.00	88.08
2	1.56	88.08
3	2.25	89.70
4	2.40	89.77
5	2.76	90.20
6	2.98	90.16
7	3.12	90.10
8	3.70	90.54
9	4.43	91.00
10	5.11	91.23
11	5.53	91.25
12	5.90	91.20
13	6.47	91.11
14	7.36	91.30
15	7.90	91.38
16	8.68	91.45
17	9.34	91.41
18	10.84	91.22
19	12.25	90.92
20	12.83	90.79

C Generator efficiency curve

The generator efficiency is provided as a function of load in MW. Table 17 lists the values used to build a piecewise–linear (PWL) approximation.

Table 17: Generator efficiency vs. load

Index	**Load [MW]**	η_{gen} [%]
1	0	96.90
2	18	96.90
3	20	96.90
4	30	97.60
5	40	98.10
6	50	98.40
7	60	98.65
8	65	98.70
9	70	98.79
10	75	98.80
11	80	98.80
12	85	98.80
13	90	98.70

D Link to GitHub repository

The GitHub repository containing the open source file is available at: https://github.com/SigurdBrensholm/Model_code_mid_term_hydropower_model.git Power plant specific information, in addition to prices and volumes are excluded from the file.